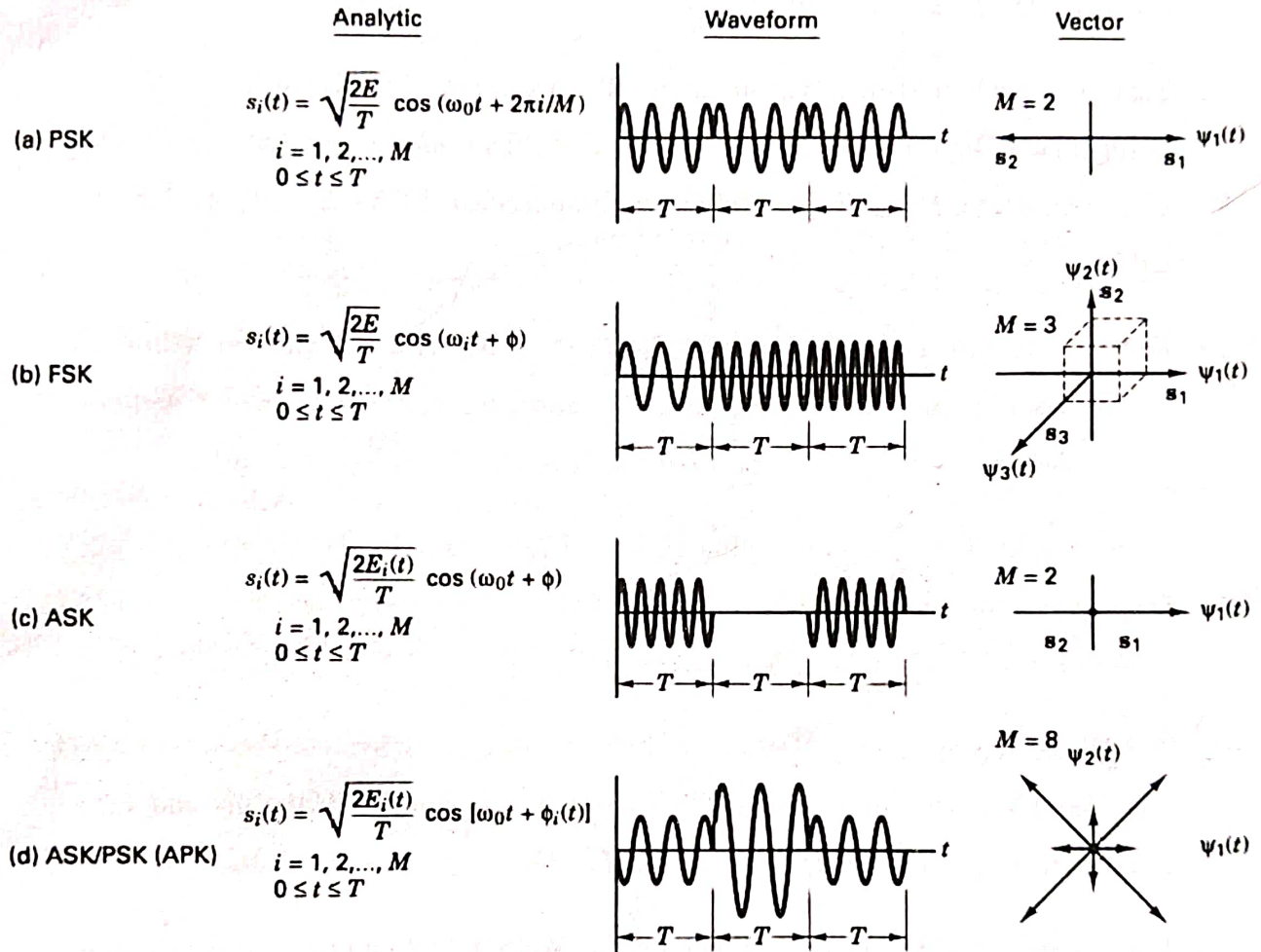


BAND PASS MODULATION

Digital modulation is the process by which digital symbols are transformed into waveforms that are compatible with the characteristics of the channel. In the case of baseband modulation, these waveforms usually take the form of shaped pulses. But in the case of *bandpass modulation* the shaped pulses modulate a sinusoid called a *carrier wave*, or simply a *carrier*; for radio transmission the carrier is converted to an electromagnetic (EM) field for propagation to the desired destination.



➤ Bit stream to symbol stream conversion

➤ Utilizes mapping a group of bits to unique values of **amplitude, phase, frequency** or combination of three parameters

Classes of Modulation

Scheme without memory
(no signature of previous symbols)

- Line codes: Polar, Unipolar, Manchester
- M-PSK, M-FSK, M-QAM

Scheme with memory
(signature of previous symbols)

- AMI and its derivatives
- DPSK

Band-limited Modulation

- Increase rate of mapping (more bit rate)
- Employ only **amplitude & phase** not frequency
- BW kept constant, power is variable and increased to maintain QoS
- Coherently demodulated
- Ex.: PAM, PSK, QAM, MSK

Power-limited Modulation

- Increase rate of mapping (more bit rate)
- Employ only **frequency & phase** not amp.
- Amplifier is at saturation (full efficiency) at the price of more BW
- Power kept constant, BW is variable and increased to maintain QoS
- Demodulated coherently or incoherently
- Ex.: Orthogonal signaling schemes, MFSK, MSK, DPSK

Any modem or mobile need both types of modulation schemes

MODULATION - BANDPASS

$$s_m(t) = \operatorname{Re} [s_{mL}(t) e^{j2\pi f_0 t}] \quad \text{over } (0, T) \\ = \operatorname{Re} [A_m g(t) e^{j2\pi f_0 t}] \quad \text{over } (0, T)$$

A_m is a Real / Complex no.

$g(t)$ is low pass shaped pulse (Real)

$e^{j2\pi f_0 t}$ is carrier wave

The values and expressions for A_m are chosen as to realize varieties of symbols that are functions of Amplitude, phase and frequency.

1. Set $f_0 = 0$: All symbols are low pass

A_m is real, PAM cases of Baseband modulation

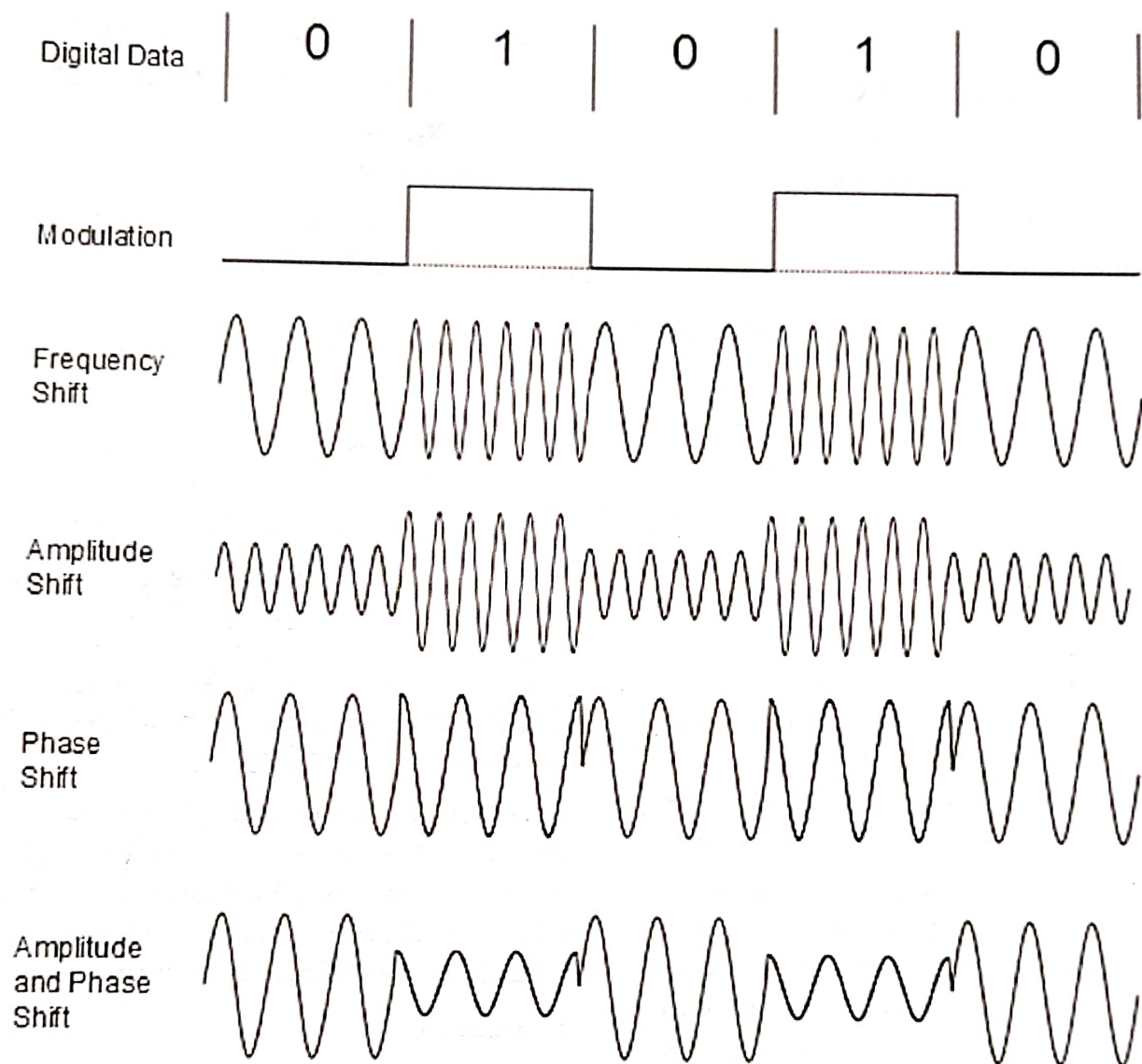
2. Set $f_0 \neq 0$ i.e. f_0 is finite bandpass center frequency.

A_m is real, - Carrier modulated PAM
- ASK.

3. Set $f_0 \neq 0$ A_m is complex, $A_m = e^{j2\pi \frac{(m-1)}{M}}$ $m=1, 2, \dots, M$
MPSK.

4. Set $f_0 \neq 0$. A_m is complex $A_m = A_m^{(r)} + j A_m^{(i)}$; $A_m^{(r)}, A_m^{(i)} \rightarrow \text{real}$
 $m=1, 2, \dots, M$.
MQAM

5. Set $f_0 \neq 0$ A_m is complex $A_m = e^{j2\pi m \Delta f t}$ $m=1, 2, \dots, M$
MFSK.



EXAMPLES OF VARIOUS SCHEMES.

10 Study these parameters

1. Define symbols mathematically. in terms of Amplitude, phase, frequency
2. Obtain basis functions
3. Represent symbols in terms of basis fun.
4. Tx and Rx block diagram → Unit 2
5. d_{ik} general formula; and $d_{min} = (d_{ik})_{min}$
6. Symbol and Average symbol energy
7. Symbol energy in terms of d_{min} / d_{min} in terms of Symbol energy.
8. Obtain PSD
9. Expression for P_e → Unit 2

LOW PASS - BAND PASS BASIS FUNCTION

Band pass $s_m(t) = \text{Re} [s_{mL}(t) e^{j2\pi f_0 t}]$

$$s_{mL}(t) = \sum_{n=1}^N s_{mLn} \phi_{nL}(t) \quad m=1, \dots, M$$

1. Determine ϕ_{nL} $n=1, 2, \dots, N$ for s_{mL} $m=1, 2, \dots, M$
Low pass basis functions by Gram Schmidt procedure
2. The set of functions $\phi_n(t) = \sqrt{2} \text{Re} [\phi_{nL}(t) e^{j2\pi f_0 t}]$
is not necessarily orthonormal basis function to bandpass $s_m(t)$
3. Write $\phi_{nL}(t) = \phi_{ni}(t) + j\phi_{nq}(t)$ low pass is in general complex form
$$\begin{aligned} \phi_{nL}(t) e^{j2\pi f_0 t} &= [\phi_{ni}(t) + j\phi_{nq}(t)] e^{j2\pi f_0 t} \\ &= [\phi_{ni}(t) \cos 2\pi f_0 t - \phi_{nq}(t) \sin 2\pi f_0 t] \\ &\quad + j[\phi_{ni}(t) \sin 2\pi f_0 t + \phi_{nq}(t) \cos 2\pi f_0 t] \\ &= \frac{\phi_n(t)}{\sqrt{2}} - j \frac{\tilde{\phi}_n(t)}{\sqrt{2}} \quad \frac{1}{\sqrt{2}} \text{ is normalizing factor} \end{aligned}$$

$$\therefore \phi_n(t) = \sqrt{2} \text{Re} \{ \phi_{nL}(t) e^{j2\pi f_0 t} \}$$

$$\tilde{\phi}_n(t) = -\sqrt{2} \text{Im} \{ \phi_{nL}(t) e^{j2\pi f_0 t} \}$$

Relation to move
from LP basis
to BP basis fns

1 low pass basis functions spawns to two Bandpass
basis function

BINARY PHASE SHIFT KEYING.

$$s_m(t) = \text{Re} \left\{ A_m g(t) e^{j2\pi f_0 t} \right\} \quad (0 \text{ to } T_b)$$

Take $g(t) = \sqrt{\frac{2E_b}{T_b}}$

$$A_m = e^{j2\pi \frac{(m-1)}{M}}$$

$m=1, 2$
 $M=2$

$$A_1 = e^{j0} = 1$$

$$A_2 = e^{j2\pi \frac{1}{2}} = e^{j\pi} = -1$$

- $s_1(t) = \sqrt{\frac{2E_b}{T_b}} \cos 2\pi f_0 t$ rep symbol 1
- $s_2(t) = -\sqrt{\frac{2E_b}{T_b}} \cos 2\pi f_0 t$ rep symbol 0.

- Energy in lowpass $E_{SL} = \int_0^{T_b} \left(\sqrt{\frac{2E_b}{T_b}} \right)^2 dt = \int_0^{T_b} 2E_b dt = 2E_b$
Only 1 basis function in [LOW PASS BASIS FUNCTION]

$$\therefore \phi_{1L}(t) = \frac{s_1(t)}{\sqrt{E_{SL}}} = \frac{\sqrt{\frac{2E_b}{T_b}}}{\sqrt{2E_b}} = \sqrt{\frac{1}{T_b}}$$

This case $\phi_{1L}(t)$ is independent of 'm' so only 1 basis fcn.

- $\phi(t) = \sqrt{2} \text{Re} \left\{ \phi_{1L} e^{j2\pi f_0 t} \right\}$
 $= \sqrt{\frac{2}{T_b}} \cos 2\pi f_0 t$
- $\tilde{\phi}(t) = \sqrt{2} \text{Im} \left\{ \phi_{1L} e^{j2\pi f_0 t} \right\}$
 $= -\sqrt{\frac{2}{T_b}} \sin 2\pi f_0 t$

TWO BANDPASS BASIS FUNCTION

- $s_1(t) = \sqrt{E_b} \phi(t)$
 $s_2(t) = -\sqrt{E_b} \phi(t)$
- IN BANDPASS MODULATION
- Here also only '1' basis fcn required.

- Energy symbol $E_s = \int_0^{T_b} \left(\sqrt{\frac{2E_b}{T_b}} \cos 2\pi f_0 t \right)^2 dt = \frac{2E_b}{T_b} \int_0^{T_b} \frac{1 + \cos 4\pi f_0 t}{2} dt$

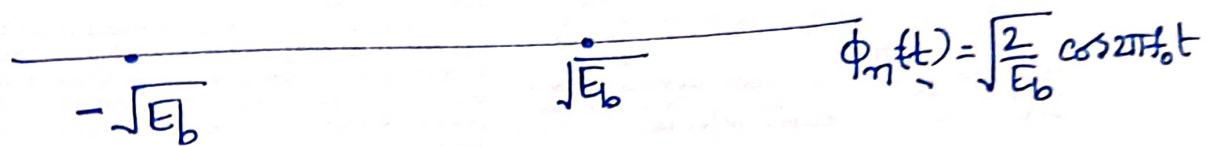
$$E_s = E_b$$

- Average symbol Energy $= \frac{1}{2} \sum_{m=1}^2 E_b = \frac{E_b + E_b}{2} = E_b$

$$E_{avg} = E_b.$$

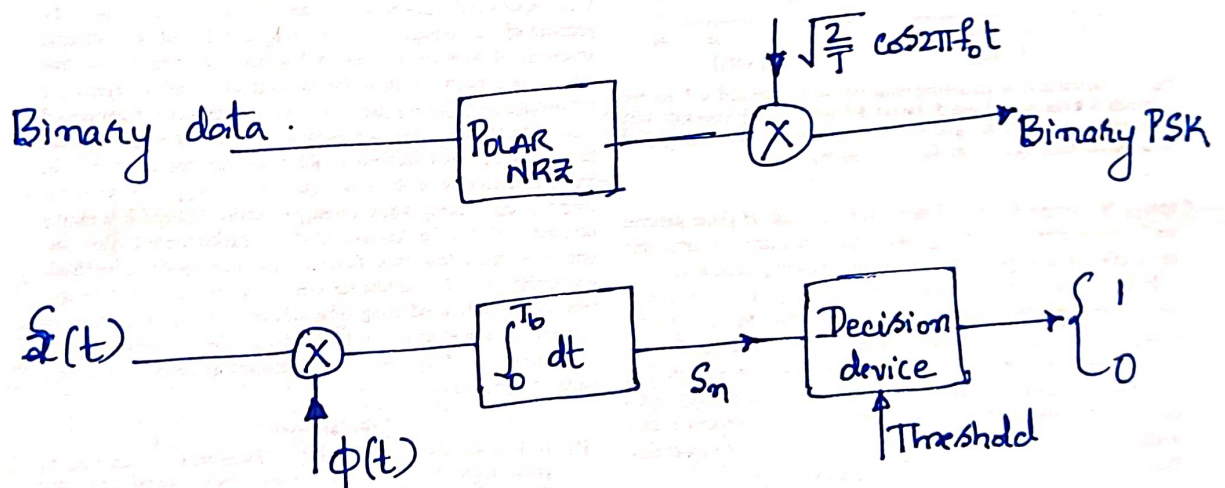
- Average bit Energy = $\frac{E_{avg}}{\log_2 2} = E_b$.

- Euclidean space



- $d_{min} = 2\sqrt{E_b}$.

- GENERATION OF COHERENT BPSK SIGNAL



- DETECTION OF COHERENT BPSK

- POWER SPECTRAL DENSITY (PSD)

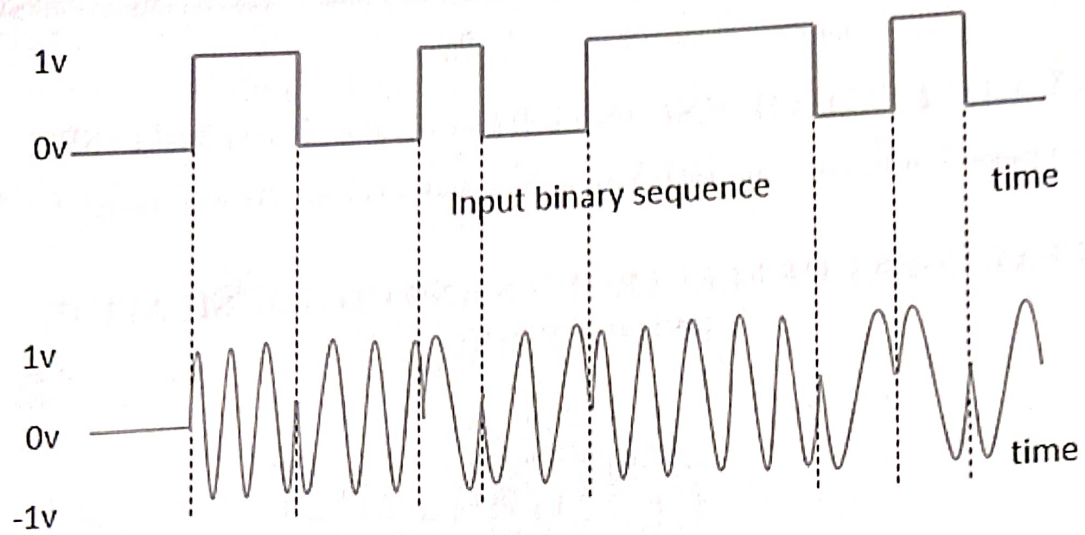
$$S(f) = \frac{|G(f)|^2}{T_b} \sum_{k=-\infty}^{\infty} R(k) e^{jk\omega T_b}$$

$$R(0) = \frac{1}{2}(A)(A) + \frac{1}{2}(-A)(-A) = A^2 = \frac{2E_b}{T_b}$$

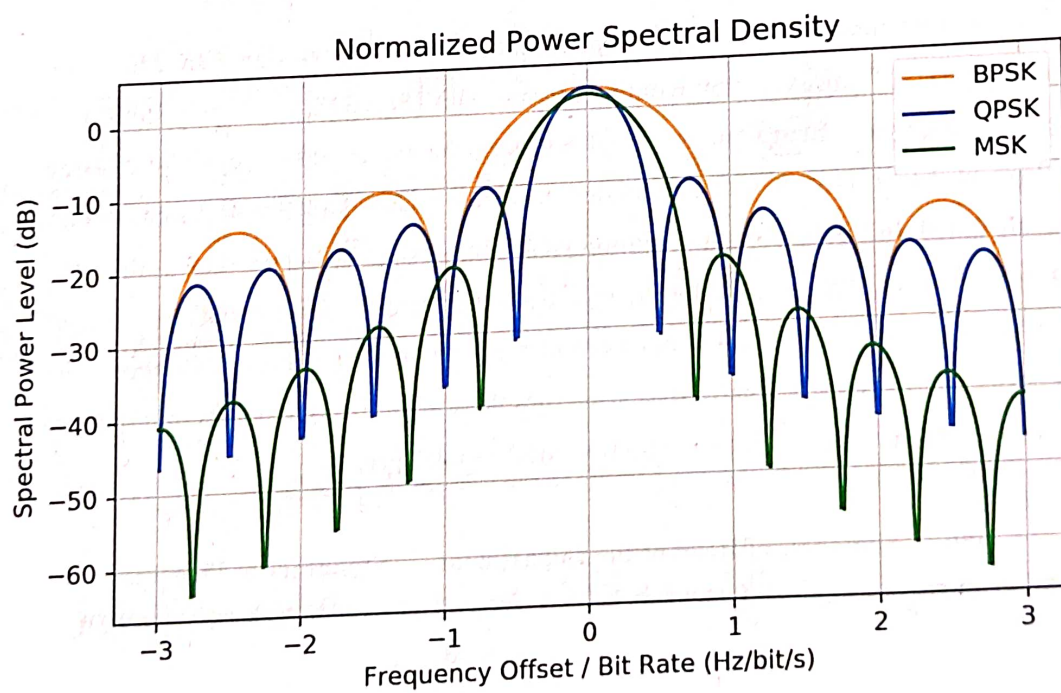
$$g(t) = \text{rect}(t/T_b) \quad G(f) = T_b \text{sinc}(fT_b)$$

$$S(f) = \frac{2E_b}{T_b} \frac{T_b^2 \text{sinc}^2(fT_b)}{T_b} = 2E_b \text{sinc}^2(fT_b)$$

$$S(f) = 2E_b \text{sinc}^2(fT_b)$$



BPSK Modulated output wave



QUADRATURE PHASE SHIFTING

$$s(t) = \text{Re} \{ A_m g(t) e^{j2\pi f_0 t} \} \quad \text{otoT}$$

$$= \text{Re} \{ s_{mL} e^{j2\pi f_0 t} \} = \text{Re} \left\{ \sum_{n=1}^M s_{mLn} \phi_{nL} e^{j2\pi f_0 t} \right\}$$

For QPSK. $A_m = e^{j(2m-1)\frac{\pi}{4}} \quad m=1,2,3,4 \quad M=4$

$$g(t) = \sqrt{\frac{2E_b}{T_b}}$$

$$s_{mL} = A_m g(t) = e^{j(2m-1)\frac{\pi}{4}} \sqrt{\frac{2E_b}{T_b}}$$

$$E_{s_{mL}} = \int_0^{T_b} A_m^2 \left(\sqrt{\frac{2E_b}{T_b}} \right)^2 dt = A_m^2 2E_b$$

$$\phi_{1L}(t) = \frac{A_m \sqrt{2E_b/T_b}}{A_m \sqrt{2E_b}} = \sqrt{1/T_b} \quad \text{There is only one low pass basis function}$$

Band pass BASIS FUNCTION.

$$\phi_n(t) = \sqrt{2} \text{Re} \{ \phi_{1L} e^{j2\pi f_0 t} \} = \sqrt{\frac{2}{T_b}} \cos 2\pi f_0 t$$

$$\tilde{\phi}_n(t) = -\sqrt{2} \text{Im} \{ \phi_{1L} e^{j2\pi f_0 t} \} = -\sqrt{\frac{2}{T_b}} \sin 2\pi f_0 t$$

$$s_m(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_0 t + (2m-1)\frac{\pi}{4}) = \sqrt{E_b} \cos(2m-1)\frac{\pi}{4} \phi(t) + \sqrt{E_b} \sin(2m-1)\frac{\pi}{4} \tilde{\phi}(t)$$

$$= \sqrt{\frac{2E_b}{T_b}} \cos(2m-1)\frac{\pi}{4} \cos 2\pi f_0 t - \sqrt{\frac{2E_b}{T_b}} \sin(2m-1)\frac{\pi}{4} \sin 2\pi f_0 t$$

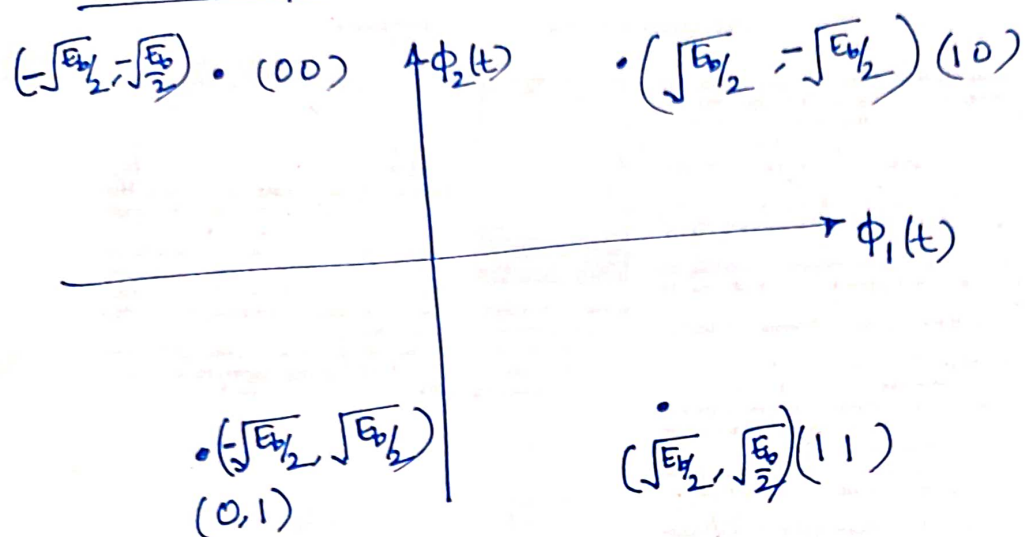
$$s_1(t) = \sqrt{E_b/2} \phi(t) + \sqrt{E_b/2} \tilde{\phi}(t) \quad \left(\sqrt{E_b/2}, \sqrt{E_b/2} \right) (1,1) \quad e^{j\pi/4}$$

$$s_2(t) = -\sqrt{E_b/2} \phi(t) + \sqrt{E_b/2} \tilde{\phi}(t) \quad \left(-\sqrt{E_b/2}, \sqrt{E_b/2} \right) (0,1) \quad e^{j3\pi/4}$$

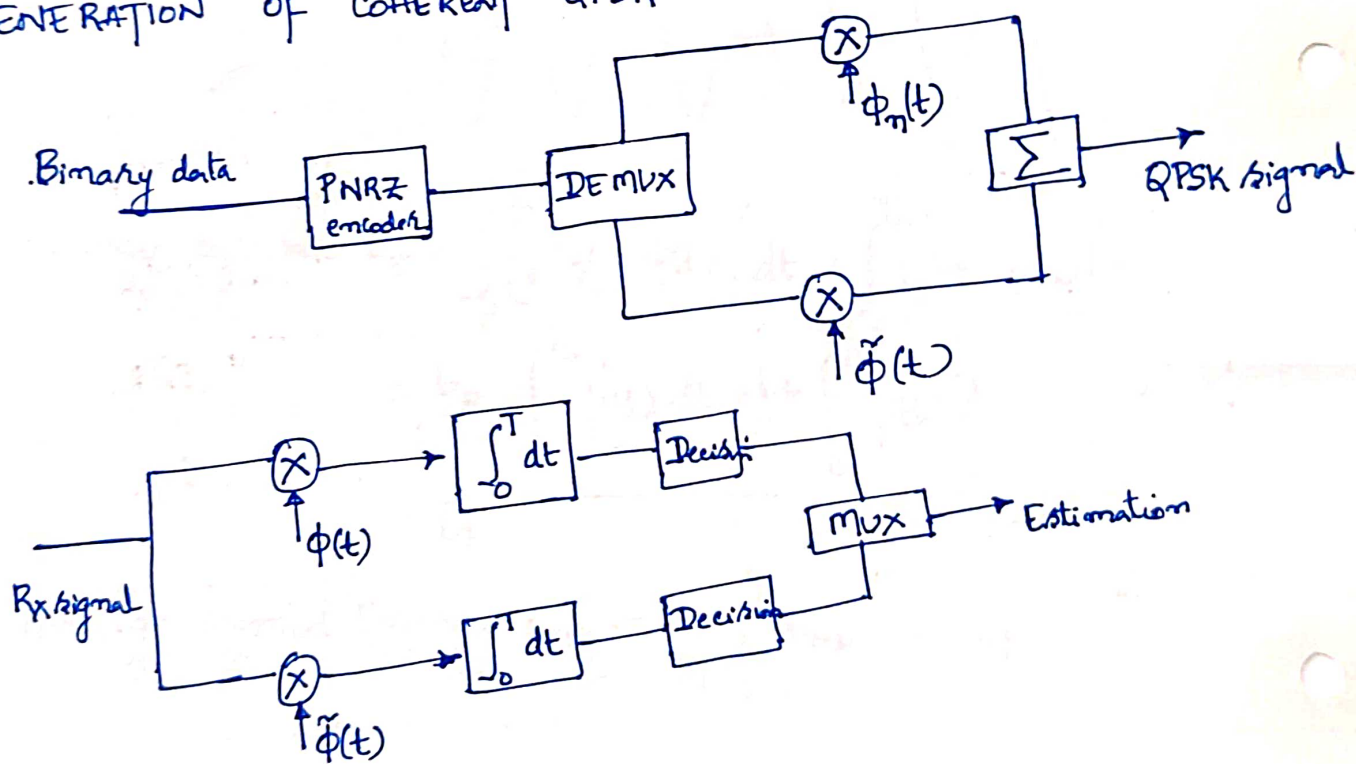
$$s_3(t) = -\sqrt{E_b/2} \phi(t) - \sqrt{E_b/2} \tilde{\phi}(t) \quad \left(-\sqrt{E_b/2}, -\sqrt{E_b/2} \right) (0,0) \quad e^{j5\pi/4}$$

$$s_4(t) = \sqrt{E_b/2} \phi(t) - \sqrt{E_b/2} \tilde{\phi}(t) \quad \left(\sqrt{E_b/2}, -\sqrt{E_b/2} \right) (1,0) \quad e^{j\pi/4}$$

EUCLIDEAN SPACE



GENERATION OF COHERENT QPSK SIGNAL



• POWER SPECTRAL DENSITY (PSD)

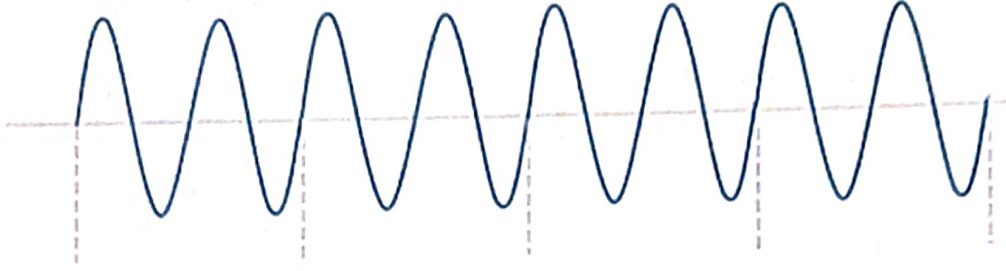
For Inphase and Quadrature phase the power spectral density is $2E_b \text{sinc}^2(fT_b)$

The base band power spectral density is

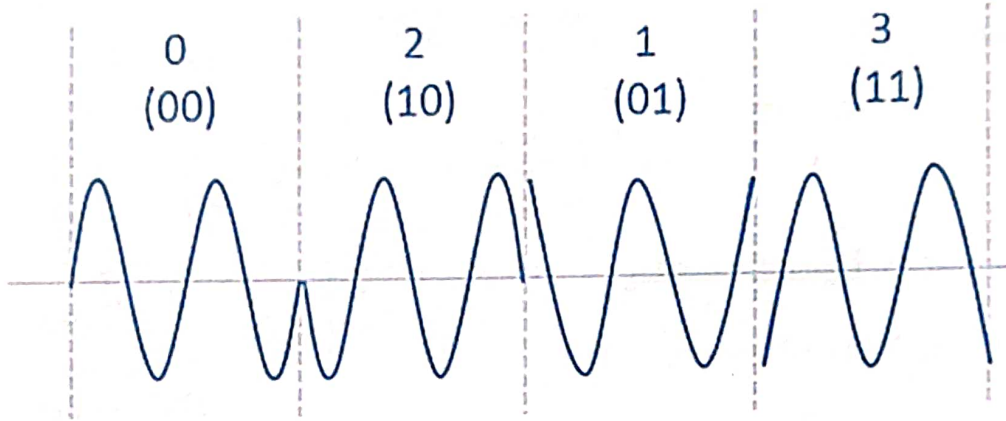
$$S(f) = 2(2E_b) \text{sinc}^2(f 2T_b)$$

$$= 4E_b \text{sinc}^2(2fT_b)$$

Carrier Signal



Modulated Signal



$$\begin{aligned}
 \text{Energy Symbol } E_s &= \int_0^{T_b} \left[\sqrt{\frac{E_b}{2}} \phi(t) \right]^2 dt + \int_0^{T_b} \left[\sqrt{\frac{E_b}{2}} \tilde{\phi}(t) \right]^2 dt \\
 &= \frac{E_b}{2} \int_0^{T_b} \phi^2(t) dt + \frac{E_b}{2} \int_0^{T_b} \tilde{\phi}^2(t) dt \\
 &= E_b
 \end{aligned}$$

$$\text{Average Symbol Energy } E_{avg} = \frac{1}{M} \sum_{i=1}^M E_{bi} = E_b$$

$$\text{Average bit Energy} = \frac{E_{avg}}{\log_2 M} = \frac{E_b}{2}$$

M-ARY PHASE SHIFT KEYING

$$s_m(t) = \operatorname{Re} \{ A_m g(t) e^{j 2\pi f_0 t} \}$$

$$= \operatorname{Re} \{ s_{mL} e^{j 2\pi f_0 t} \} = \operatorname{Re} \left\{ \sum s_{mLm} \phi_{mL} e^{j 2\pi f_0 t} \right\}$$

Here $A_m = e^{j 2\pi \frac{(m-1)}{M}}$

$$g(t) = \sqrt{\frac{2E_b}{T_b}}$$

$$s_{mL} = A_m g(t) = e^{j 2\pi \frac{(m-1)}{M}} \sqrt{\frac{2E_b}{T_b}}$$

$$E_{s_{mL}} = \int_0^{T_b} A_m^2 \left(\sqrt{\frac{2E_b}{T_b}} \right)^2 dt = A_m^2 2E_b$$

$$\phi_L = \frac{A_m \sqrt{\frac{2E_b}{T_b}}}{A_m \sqrt{2E_b}} = \sqrt{\frac{1}{T_b}}$$

Since ϕ_L is not dependent on m , there is only 1 low pass basis function

Band pass basis function

$$\phi(t) = \sqrt{\frac{2}{T_b}} \cos 2\pi f_0 t$$

$$\tilde{\phi}(t) = -\sqrt{\frac{2}{T_b}} \sin 2\pi f_0 t$$

$$s_m(t) = \sqrt{\frac{2E_b}{T_b}} \cos \left(2\pi f_0 t + \frac{2\pi(m-1)}{M} \right)$$

$$= \sqrt{\frac{2E_b}{T_b}} \cos \frac{2\pi}{M} (m-1) \cos 2\pi f_0 t - \sqrt{\frac{2E_b}{T_b}} \sin \frac{2\pi}{M} (m-1) \sin 2\pi f_0 t$$

$$\therefore s_m(t) = \sqrt{E_b} \cos \frac{2\pi}{M} (m-1) \phi(t) - \sqrt{E_b} \sin \frac{2\pi}{M} (m-1) \tilde{\phi}(t)$$



Points $\left(\sqrt{E_b} \cos \frac{2\pi(m-1)}{M}, \sqrt{E_b} \sin \frac{2\pi(m-1)}{M} \right)$

Distance between two points in Euclidean space

$$d_{mn} = \sqrt{|\mathbf{s}_m - \mathbf{s}_n|^2}$$

$$\mathbf{s}_m = \left[\sqrt{E_b} \cos \frac{2\pi(m-1)}{M}, \sqrt{E_b} \sin \frac{2\pi(m-1)}{M} \right]$$

$$\mathbf{s}_n = \left[\sqrt{E_b} \cos \frac{2\pi(n-1)}{M}, \sqrt{E_b} \sin \frac{2\pi(n-1)}{M} \right]$$

$$d_{mn}^2 = \left\{ E_b \cos^2 \frac{2\pi(m-1)}{M} + E_b \cos^2 \frac{2\pi(n-1)}{M} - 2E_b \cos \frac{2\pi(m-1)}{M} \cos \frac{2\pi(n-1)}{M} + \right. \\ \left. E_b \sin^2 \frac{2\pi(m-1)}{M} + E_b \sin^2 \frac{2\pi(n-1)}{M} - 2E_b \sin \frac{2\pi(m-1)}{M} \sin \frac{2\pi(n-1)}{M} \right\}$$

$$= 2E_b [1 - \cos \frac{2\pi(m-n)}{M}] = 2E_b \left[1 - \cos \frac{2\pi(m-n)}{M} \right]$$

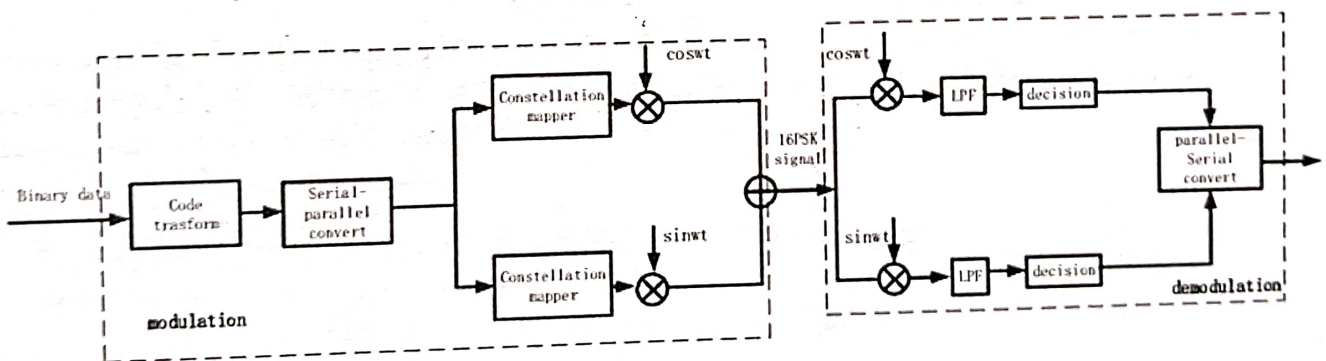
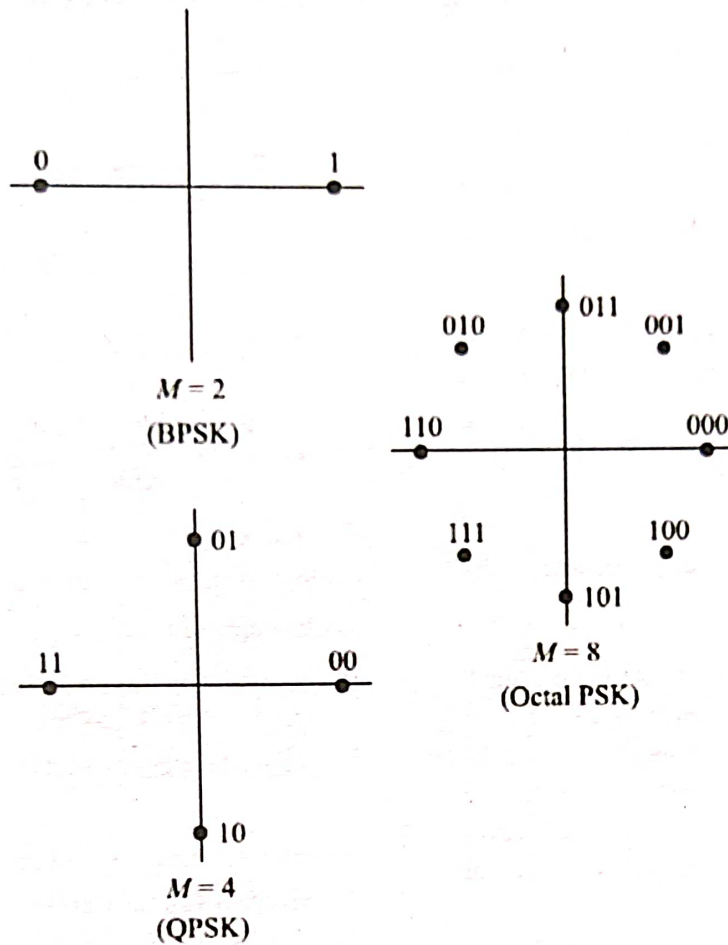
$$= 2E_b (2 \sin^2 \frac{\pi(m-n)}{M})$$

$$1 - \cos^2 \theta = 2 \sin^2 \frac{\theta}{2}$$

$$\therefore d_{mn} = 2 \sqrt{E_b} \sin \frac{\pi(m-n)}{M}$$

$$d_{min} = 2 \sqrt{E_b} \sin \frac{\pi}{M}$$

- Energy symbol = E_b
- Average symbol energy = $\frac{1}{M} \sum E_b = E_b$
- Average $E_{\text{bit}} = \frac{E_b}{\log_2 M}$
- Symbol energy $E_b = K E_{\text{bit av}} = \log_2 M E_{\text{bit av}}$
- $d_{min} = 2 \sqrt{\log_2 M E_{\text{bit av}}} \cdot \sin \frac{\pi}{M}$



PSD of M-ary PSK

Symbol duration $T = T_b \log_2 M$

Av. bit Energy $= E_b \log_2 M$

$$S(f) = 2E \operatorname{sinc}^2(Tf)$$

$$= 2(E_b \log_2 M) \operatorname{sinc}^2(T_b \log_2 M f)$$

(BASE BAND)

PULSE AMPLITUDE MODULATION

The signal waveform is given by
 $s_m(t) = A_m p(t) \quad 1 \leq m \leq M$

$p(t)$ is pulse duration T

$A_m = 2m-1-m \quad m=1, 2, \dots, M ; M=2^K$ K is bit blocks of symbols.

$$A_m = \pm 1, \pm 3, \dots, \pm(m-1)$$

The energy of signal $s_m(t)$, $p(t)$ is real valued signal pulse

$$E_m = \int_{-\infty}^{\infty} A_m^2 p^2(t) dt = A_m^2 E_p \quad E_p \text{ is energy of } p(t)$$

$$\text{Energy Average } E_{avg} = \frac{E_p}{M} \sum_{m=1}^M A_m^2$$

$$= \frac{2E_p}{M} [1^2 + 3^2 + \dots + (m-1)^2]$$

$$= \frac{2E_p}{M} \frac{m(m^2-1)}{6}$$

$$E_{avg} = \text{Energy Average in PAM} = \frac{E_p}{3} (m^2-1)$$

$$\text{Average Bit Energy } E_{bavg} = \frac{E_{avg}}{\log_2 M} = \frac{(m^2-1)E_p}{3 \log_2 M}$$

NOTE

$$\therefore s_m(t) = A_m p(t)$$

$$A_m = 2m-1-m \quad m=1, 2, \dots, M$$

$$E_m = A_m^2 E_p$$

$$E_{avg} = \frac{E_p}{3} (m^2-1)$$

$$E_{bavg} = \frac{m^2-1}{3 \log_2 M} E_p$$

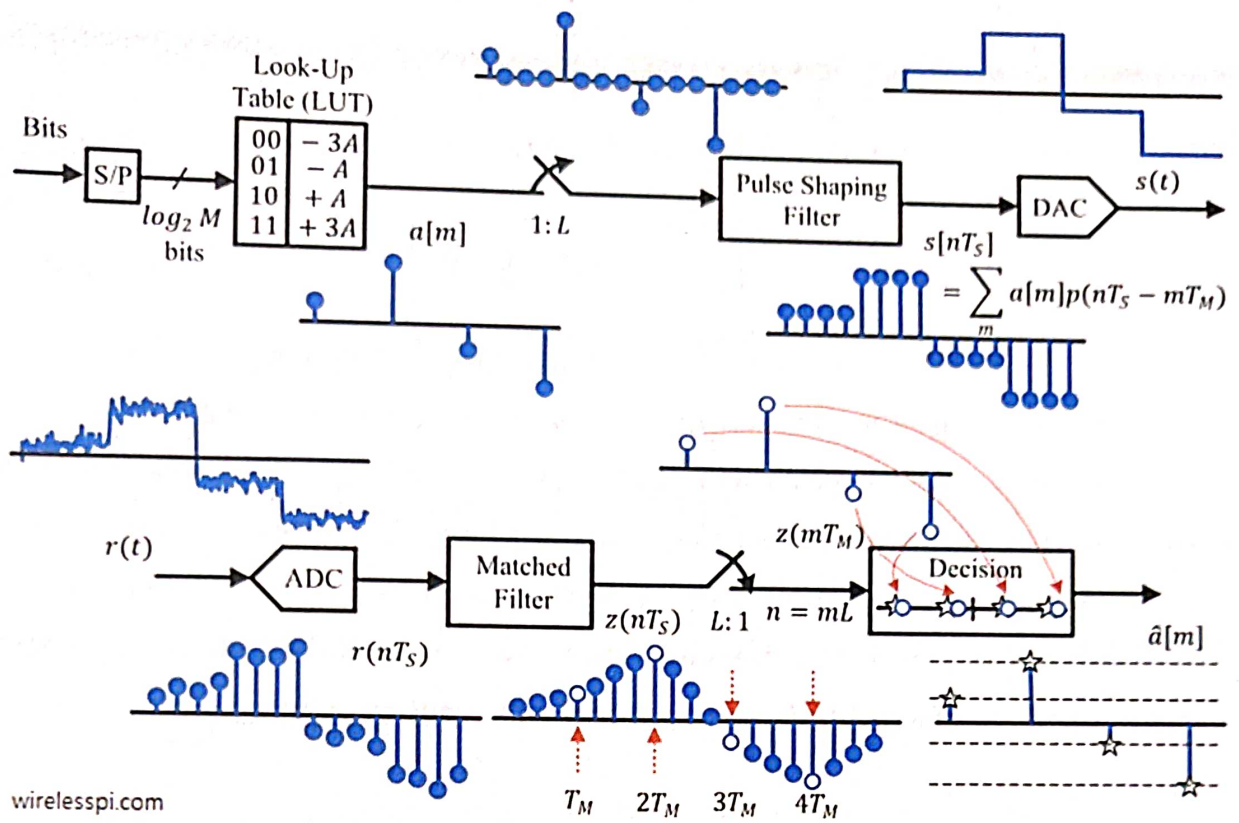


Fig 1: PAM Modulation and Demodulation

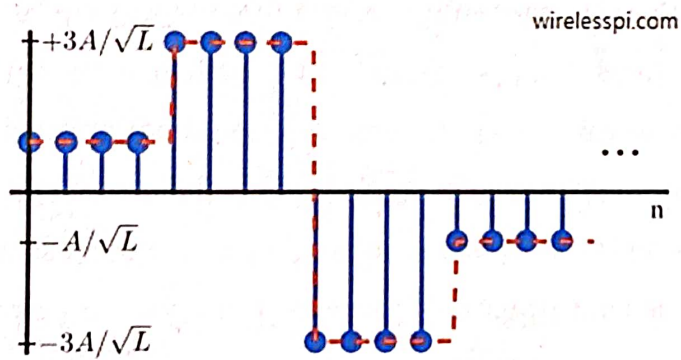


Fig. 2: 4 level PAM Signal

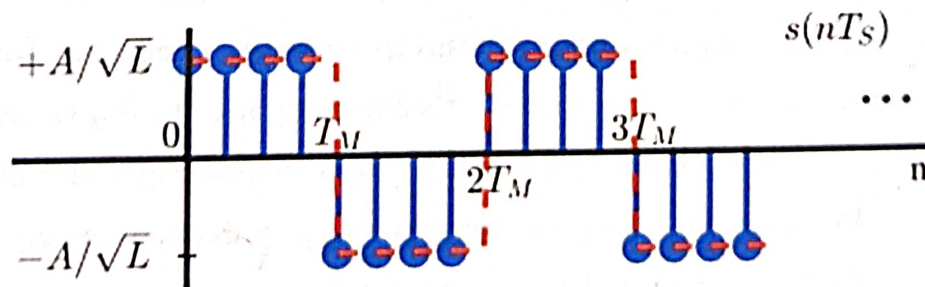


Fig. 3: 2 level PAM Signal

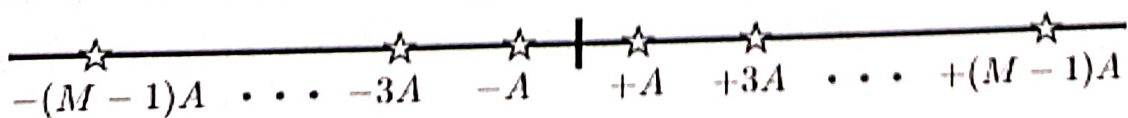


Fig. 4: Constellation Diagram

CARRIER MODULATED PAM

In many cases carrier modulated band pass signals with lowpass equivalents $A_m g(t)$ where $g(t)$ is real.

$$s_m(t) = \operatorname{Re} \{ s_{ml} e^{j2\pi f_c t} \}$$

$$= \operatorname{Re} \{ A_m g(t) e^{j2\pi f_c t} \}$$

$$= A_m g(t) \cos 2\pi f_c t. \quad f_c \text{ is carrier freq.}$$

$$\begin{aligned} \text{Energy in Carrier Modulated PAM} &= \int_0^T A_m^2 g^2(t) \cos^2 2\pi f_c t \, dt \\ E_m &= \int_0^T A_m^2 g^2(t) \frac{1 + \cos 4\pi f_c t}{2} \, dt \\ &= \int_0^T A_m^2 \frac{g^2(t)}{2} \, dt + \int_0^T A_m^2 \frac{g^2(t)}{2} \cos 4\pi f_c t \, dt \\ E_m &= A_m^2 \frac{E_g}{2} \end{aligned}$$

$$\begin{aligned} \text{Energy Average } E_{avg} &= \frac{1}{M} \frac{E_g}{2} \sum_{m=1}^M A_m^2 \\ &= \frac{E_g}{2M} [1 + 3^2 + (M-1)^2] = \\ &= \frac{E_g}{2M} \frac{M(M^2-1)}{6} = \frac{E_g (M^2-1)}{6} \end{aligned}$$

$$\text{Average bit Energy } E_{bavg} = \frac{E_g (M^2-1)}{6 \log_2 M}$$

$$s_{mL}(t) = A_m g(t)$$

and we have energy $A_m^2 E_g$

Then basis function $\phi_{mL} = \frac{s_{mL}(t)}{A_m \sqrt{E_g}} = \frac{A_m g(t)}{A_m \sqrt{E_g}}$

Here $m=1$

$$\therefore \phi_{1L} = \frac{g(t)}{\sqrt{E_g}}$$

Signal representation using basis fn is $s_{mL}(t) = A_m \sqrt{E_g} \phi_{1L}(t)$

Now convert $\phi_{1L}(t)$ into bandpass basis fn. $\{\phi_1, \tilde{\phi}_1\}$

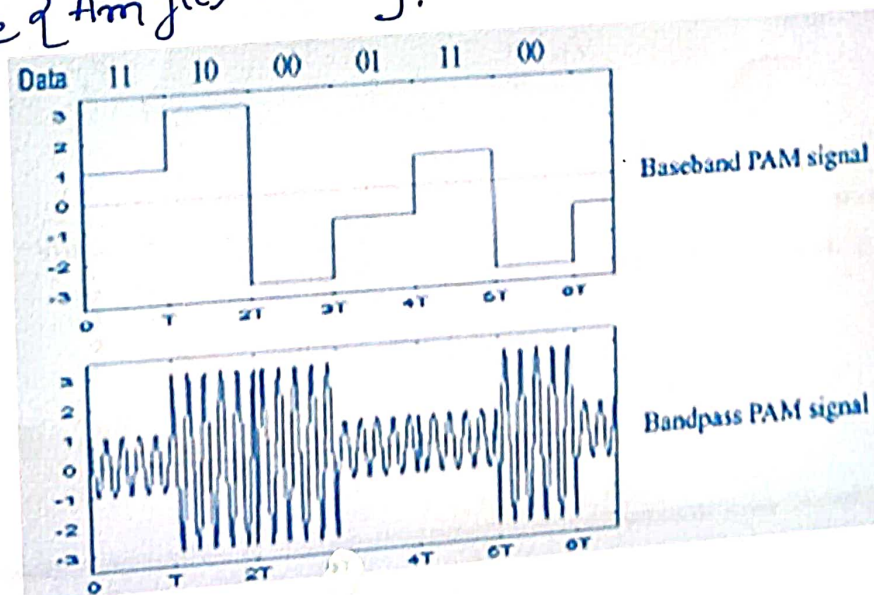
$$\phi_1 = \sqrt{2} \operatorname{Re} \left\{ \phi_{1L} e^{j2\pi f_c t} \right\} = \sqrt{2} \operatorname{Re} \left\{ \frac{g(t)}{\sqrt{E_g}} e^{j2\pi f_c t} \right\}$$

$$= \sqrt{\frac{2}{E_g}} g(t) \cos 2\pi f_c t$$

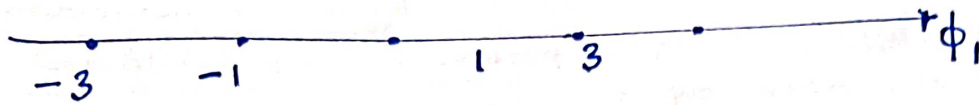
$$\tilde{\phi}_1 = -\sqrt{2} \operatorname{Im} \left\{ \phi_{1L} e^{j2\pi f_c t} \right\} = -\sqrt{\frac{2}{E_g}} g(t) \sin 2\pi f_c t$$

$$s_{mL}(t) = \frac{s_{mL}}{\sqrt{2}} \phi_1 + \frac{j s_{mL}}{\sqrt{2}} \tilde{\phi}_1 = A_m \sqrt{\frac{E_g}{2}} \phi_1 \quad \text{only 1 required}$$

$$= \operatorname{Re} \left\{ A_m g(t) e^{j2\pi f_c t} \right\}$$



Bandpass Constellation



$$d_{mn} = \sqrt{\|\tilde{s}_m - \tilde{s}_n\|^2}$$

$$d_{mn}^2 = \left[A_m \sqrt{\frac{E_g}{2}} - A_n \sqrt{\frac{E_g}{2}} \right]^2 = [A_m - A_n]^2 \frac{E_g}{2}$$

$$d_{mn} = |A_m - A_n| \sqrt{\frac{E_g}{2}}$$

$$A_m - A_n = (2m-1) - (2n-1) = 2(m-n) = 2 \text{ (minimum)}$$

$$\therefore d_{\min} = \sqrt{2E_g} \text{ in bandpass PAM}$$

$$d_{\min} = 2\sqrt{E_p} \text{ in base band ie } d_{mn} = \sqrt{(A_m - A_n)^2 E_p} = |A_m - A_n| \sqrt{E_p}$$

$$E_{\text{avg}} = \frac{E_g (m^2 - 1)}{6 \log_2 m} \quad \therefore E_g = \frac{6 \log_2 m}{m^2 - 1} E_{\text{avg}}$$

$$\therefore d_{\min} = \sqrt{\frac{12 \log_2 m}{m^2 - 1} E_{\text{avg}}}$$

QUADRATURE AMPLITUDE MODULATION

$$s_m(t) = \text{Re} [A_m g(t) e^{j2\pi f_0 t}]$$

here $A_m = a_m + j b_m$ $m = 0, \pm 1, \pm 2, \dots$

$$g(t) = \sqrt{\frac{2E_b}{T_b}}$$

$$s_m(t) = \text{Re} \left\{ (a_m + j b_m) \sqrt{\frac{2E_b}{T_b}} (\cos 2\pi f_0 t + j \sin 2\pi f_0 t) \right\}$$

$$= a_m \sqrt{\frac{2E_b}{T_b}} \cos 2\pi f_0 t - b_m \sqrt{\frac{2E_b}{T_b}} \sin 2\pi f_0 t$$

$$s_{mL}(t) = A_m g(t) = (a_m + j b_m) g \sqrt{\frac{2E_b}{T_b}}$$

$$E_{s_{mL}} = \int_0^{T_b} \sqrt{\frac{2E_b}{T_b}} (A_m)^2 dt = \int_0^{T_b} (a_m + j b_m)^2 \frac{2E_b}{T_b} dt$$

$$= (a_m + j b_m)^2 2E_b$$

LOW PASS BASIS FUNCTION

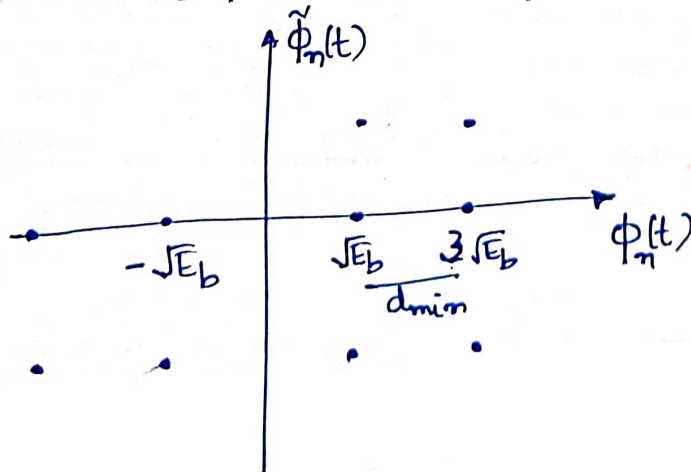
$$\phi_L(t) = \frac{s_{mL}(t)}{\sqrt{E_{s_{mL}}}} = \frac{(a_m + j b_m) \sqrt{\frac{2E_b}{T_b}}}{(a_m + j b_m) \sqrt{2E_b}} = \frac{1}{\sqrt{T_b}} \quad \text{Only 1 basis function}$$

BANDPASS BASIS FUNCTION

$$\phi_m(t) = \sqrt{2} \text{Re} \{ \phi_L e^{j2\pi f_0 t} \} = \sqrt{\frac{2}{T_b}} \cos 2\pi f_0 t$$

$$\tilde{\phi}_m(t) = -\sqrt{2} \text{Im} \{ \phi_L e^{j2\pi f_0 t} \} = -\sqrt{\frac{2}{T_b}} \sin 2\pi f_0 t$$

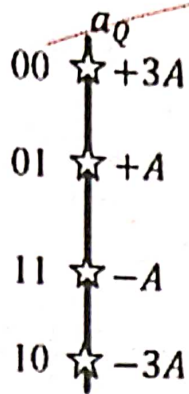
$$s_m(t) = a_m \sqrt{E_b} \phi(t) + b_m \sqrt{E_b} \tilde{\phi}(t)$$



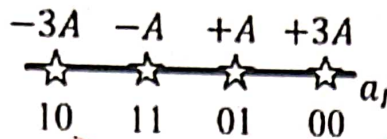
Take $a_m = (2i-1)\sqrt{E_b}$
 $b_m = (2j-1)\sqrt{E_b}$

(Any point denoted as $(a_m \sqrt{E_b}, b_m \sqrt{E_b})$)

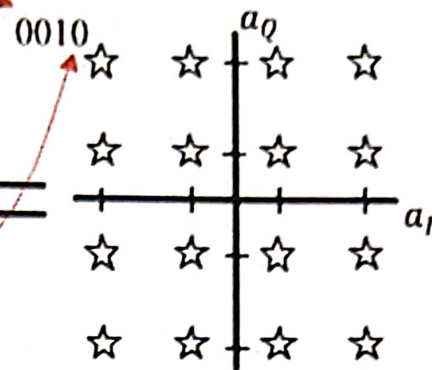
$\sqrt{M}$ -PAM



$\sqrt{M}$ -PAM

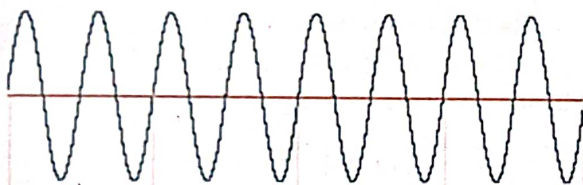


M-QAM



DIGITAL QAM (8QAM)

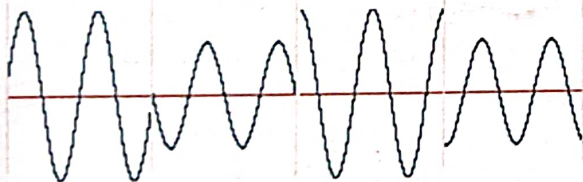
Carrier / Channel



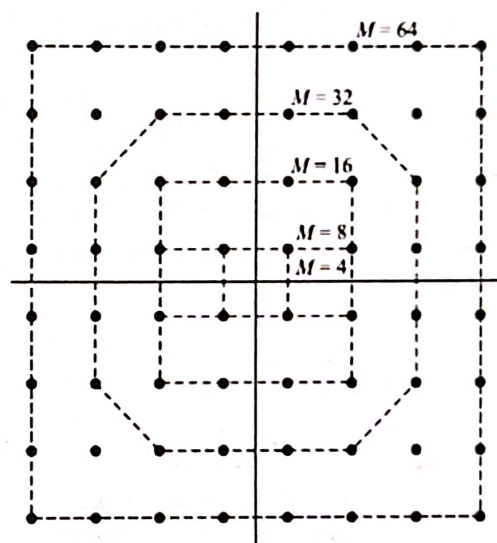
Modulating value from three bits.

0 (000) 6 (110) 1 (001) 7 (111)

Modulated Result



Note: Only four (0, 6, 1 and 7) out of the eight possible modulation states (0-7) are shown in this illustration.



M-QAM CONSTITUTION

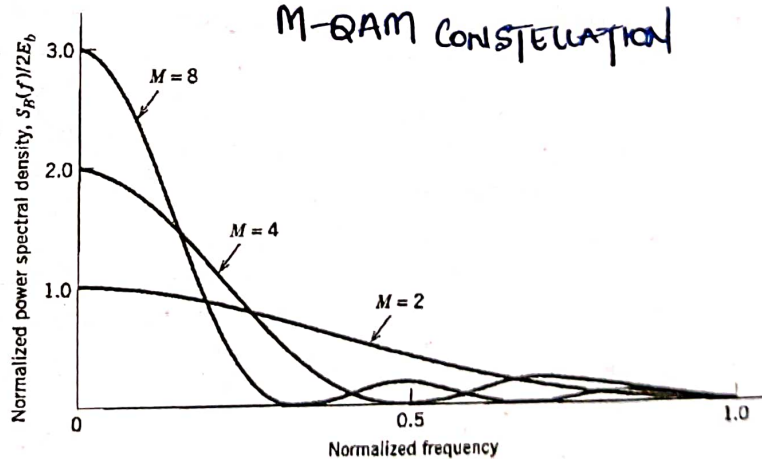
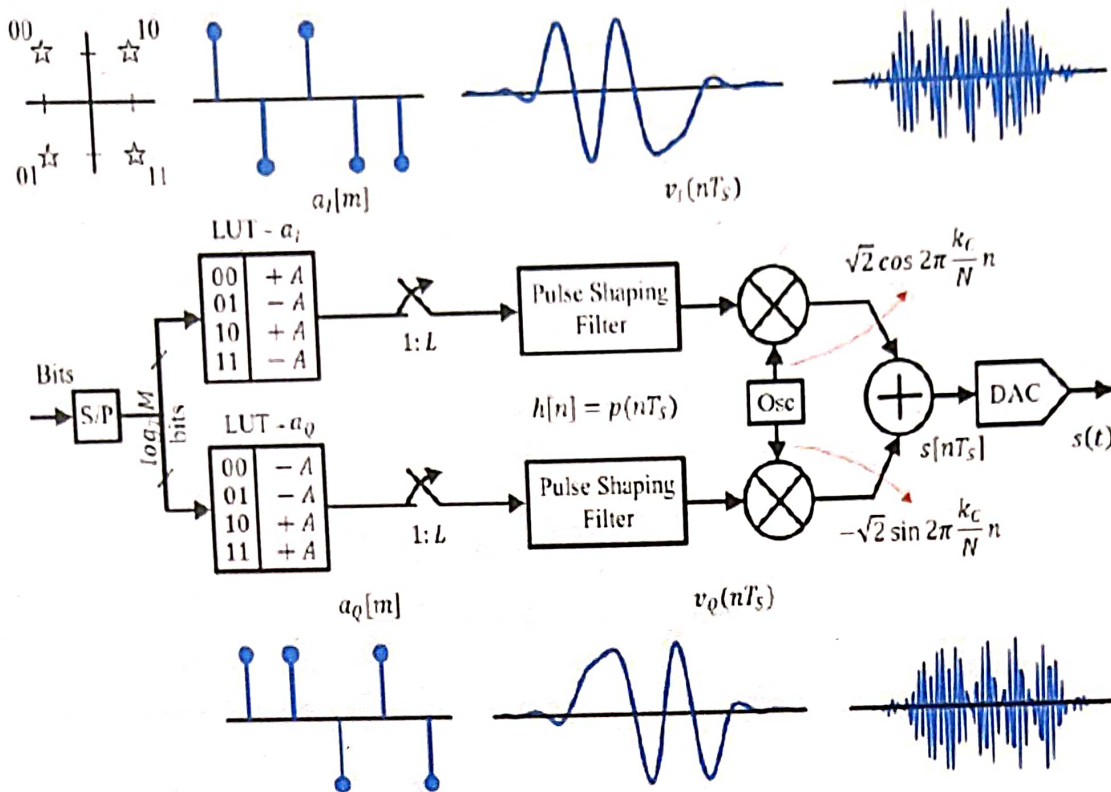
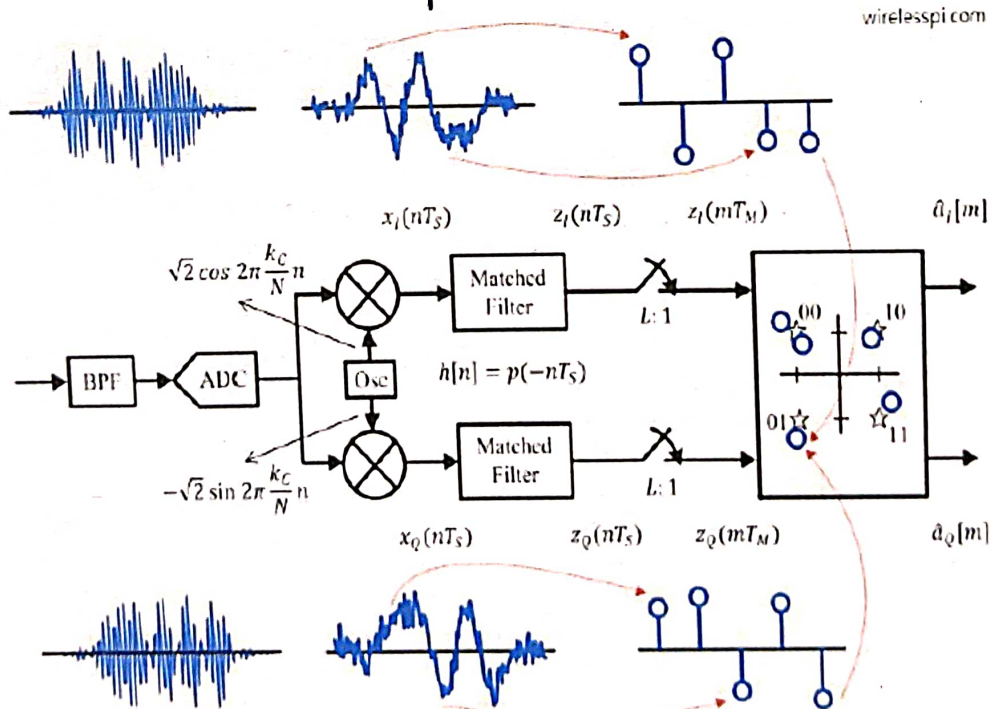


FIGURE 6.16 Power spectra of M-ary PSK signals for $M = 2, 4, 8$.

QAM - MODULATOR



QAM DEMODULATION.



$$d_{min} = 2\sqrt{E_b} \left\{ (3\sqrt{E_b} - \sqrt{E_b}) = 2\sqrt{E_b} \right\}$$

$$a_m = (2i-1)\sqrt{E_b} = (2i-1)\frac{d_{min}}{2}$$

$$b_m = (2j-1)\sqrt{E_b} = (2j-1)\frac{d_{min}}{2}$$

$$i, j = \pm 1, \pm 3, \dots \dots \pm \sqrt{M}$$

$$\text{also } \sqrt{M} = L$$

$$\text{Energy symbol } E_s = \int_0^{T_b} (a_m \sqrt{E_b})^2 \phi_m^2(t) dt + \int_0^{T_b} (b_m \sqrt{E_b})^2 \tilde{\phi}_m(t) dt$$

$$= a_m^2 E_b + b_m^2 E_b$$

$$E_s = a(2i-1)^2 E_b + (2j-1)^2 E_b$$

$$\text{Energy symbol Avg} = \frac{1}{\sqrt{M}} \sum_{i=-\sqrt{M}}^{\sqrt{M}} (a_i \sqrt{E_b})^2 + \frac{1}{\sqrt{M}} \sum_{j=-\sqrt{M}}^{\sqrt{M}} (b_j \sqrt{E_b})^2$$

$$\frac{1}{\sqrt{M}} \frac{1}{\sqrt{M}} \sum_{i=-\sqrt{M}}^{\sqrt{M}} \sum_{j=-\sqrt{M}}^{\sqrt{M}} (a_i \sqrt{E_b})^2 + (b_j \sqrt{E_b})^2$$

$$= \frac{E_b}{M} \cdot 2 \sum_{i=1}^{\sqrt{M}} \sum_{j=1}^{\sqrt{M}} a_i^2 + b_j^2$$

$$a_m = 2i-1$$

$$b_j = 2j-1$$

$$= 2 \frac{E_b}{M} \sum_{i=1}^{\sqrt{M}} \sum_{j=1}^{\sqrt{M}} (2i-1)^2 + (2j-1)^2$$

$$= 4 \frac{E_b}{M} \sum_{j=1}^{\sqrt{M}} \sum_{i=1}^{\sqrt{M}} (2i-1)^2$$

$$= 4 \frac{E_b}{M} \sum_j \frac{\sqrt{M}(\sqrt{M}^2-1)}{6} = 4 \frac{E_b}{M} \frac{\sqrt{M}(M-1)}{6} \sum_{j=1}^{\sqrt{M}} (1)$$

$$E_{\text{sym Avg}} = 4 \frac{E_b}{M} \frac{\sqrt{M}(M-1)}{6} \sqrt{M} = \frac{2}{3} E_b (M-1)$$

$$E_{\text{bit Avg}} = \frac{2}{3} \frac{E_b (M-1)}{\log_2 M}$$

Offset QUADRATURE PHASE SHIFT KEYING

In QPSK.

$$S(t) = \frac{1}{\sqrt{2}} a_1(t) \cos(2\pi f_c t + \frac{\pi}{4}) + \frac{1}{\sqrt{2}} a_2(t) \sin(2\pi f_c t + \frac{\pi}{4})$$

here $a_k = \pm 1$; $k=0,1,2,\dots$ arrive at rate $1/T$ and is separated into two streams $a_1(t)$ and $a_2(t)$.

$$S(t) = \cos[2\pi f_c t + \theta(t)] \quad \theta(t) = 0; \pm 90^\circ, 180^\circ$$

for 4 combinations $a_1(t)$ and $a_2(t)$

In QPSK the odd and even bit streams, transmitted at rate $1/2T$ baud are synchronously aligned such that their transitions coincide.

In OQPSK is obtained by shift or offset in $a_1(t)$ and $a_2(t)$ data stream by amount T .

- In QPSK, carrier phase over at any $2T$ intervals in any of 4 phases
 - If one component changes sign phase shift of $\pm 90^\circ$ occurs
 - Change in two components phase shift of 180°
- Thus QPSK will no longer have constant envelope.

In OQPSK both components can't change states simultaneously only one changes states at every T states by 0° or $\pm 90^\circ$

To reduce the amplitude fluctuations of QPSK, the bit stream responsible for generation of Quadrature component is reduced delayed by half a symbol interval with reference to inphase component

Then the basis functions are

$$\phi_1(t) = \sqrt{\frac{2}{T}} \cos 2\pi f_c t \quad 0 \leq t \leq T$$

$$\phi_2(t) = \sqrt{\frac{2}{T}} \sin 2\pi f_c t \quad \frac{T}{2} \leq t \leq \frac{3T}{2}$$

So the Signal diagram changes as shown below

- All other parameters such as Energy, dmm, & PSD is same as QPSK
- DQPSK waveforms are shown in next page

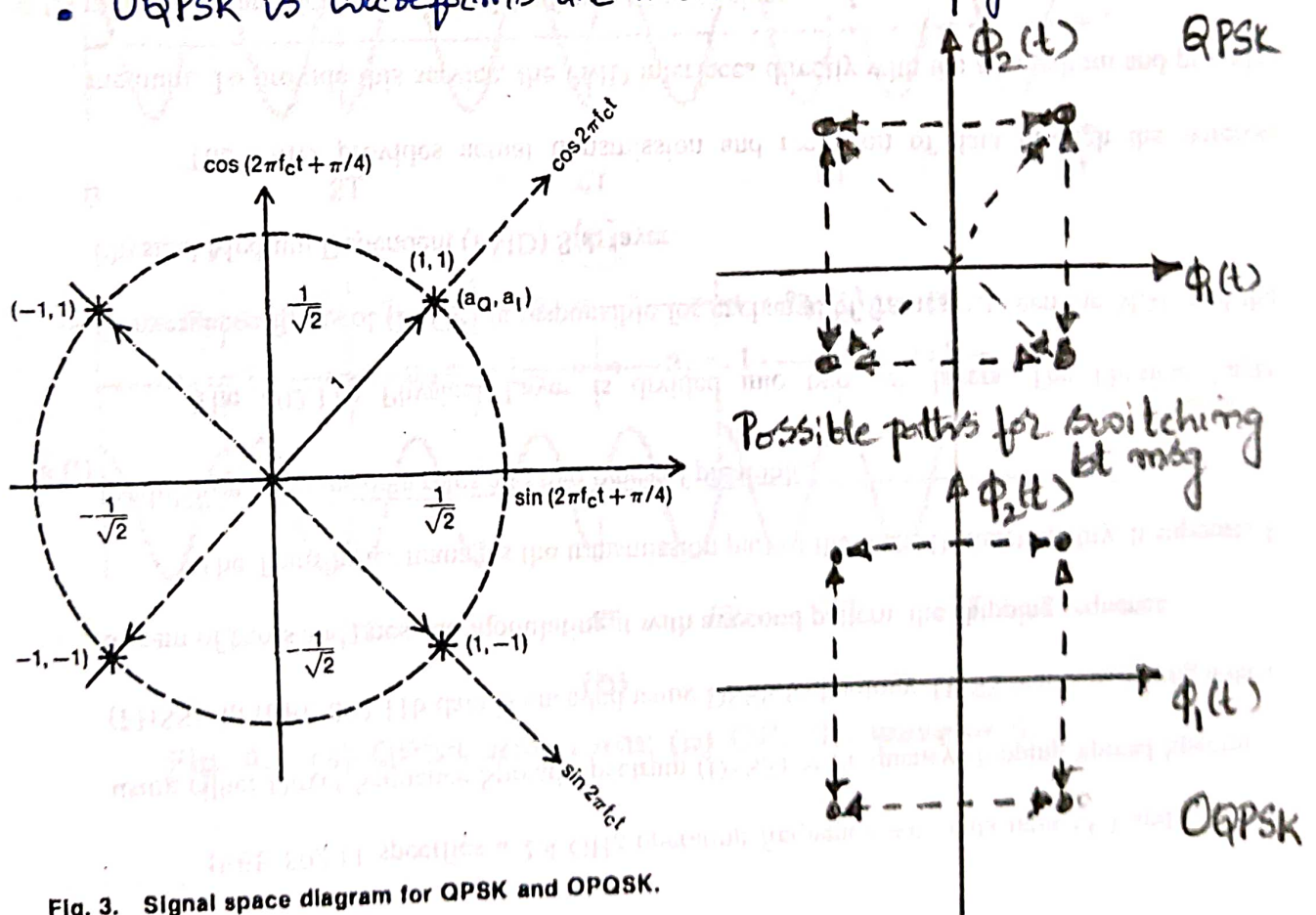


Fig. 3. Signal space diagram for QPSK and OQPSK.

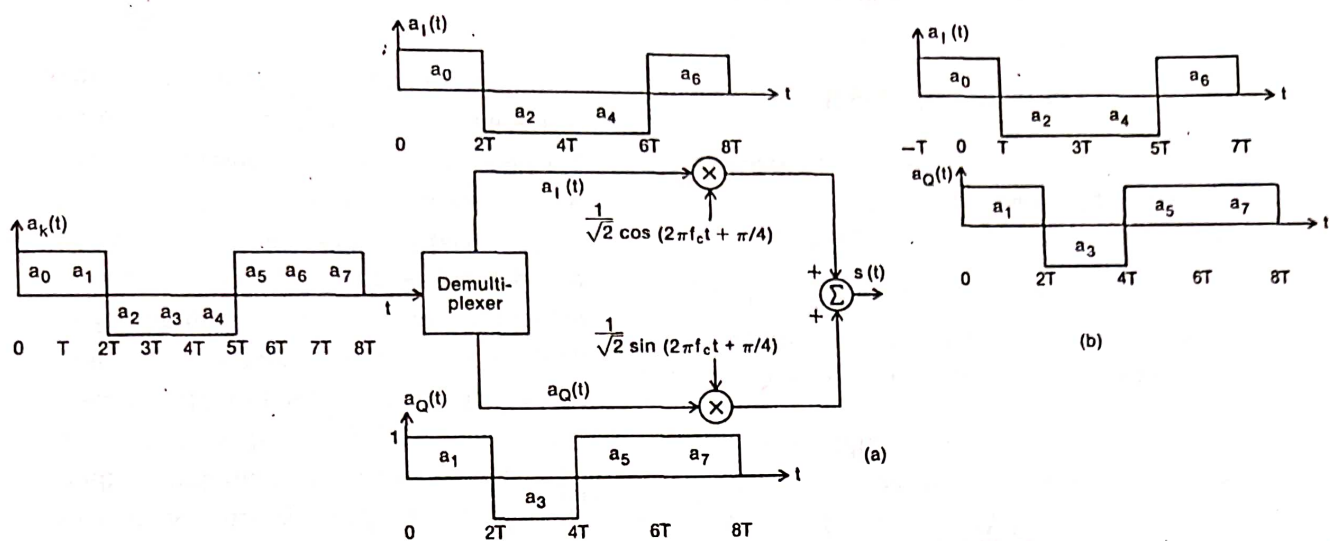


Fig. 2. (a) QPSK modulator; (b) Staggering of data streams in OPQSK.

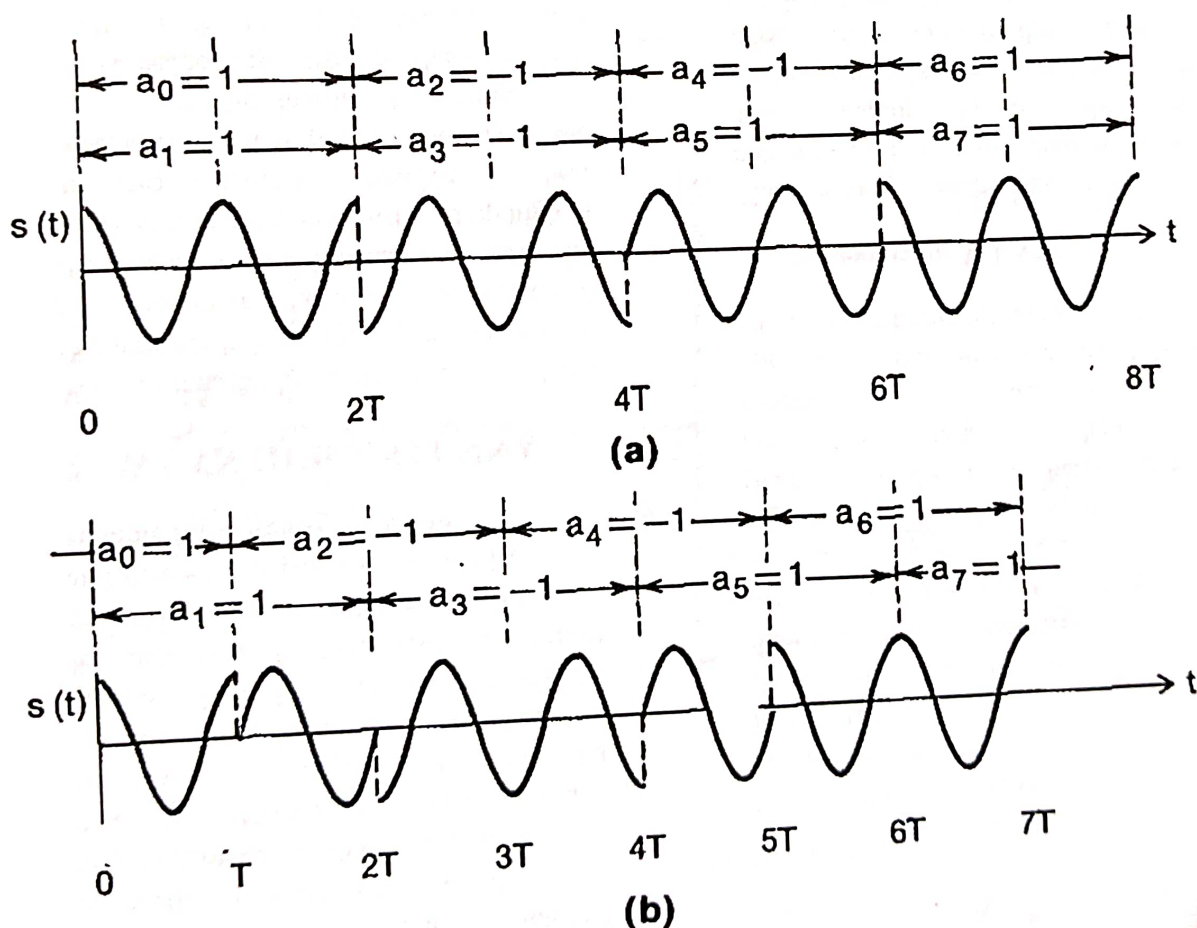


Fig. 4. (a) QPSK waveform; (b) OPQSK waveform.

ORTHOGONAL SIGNALS

Derivation not included in syllabus

If signals $s_m(t)$ and $s_n(t)$ are said to be orthogonal if

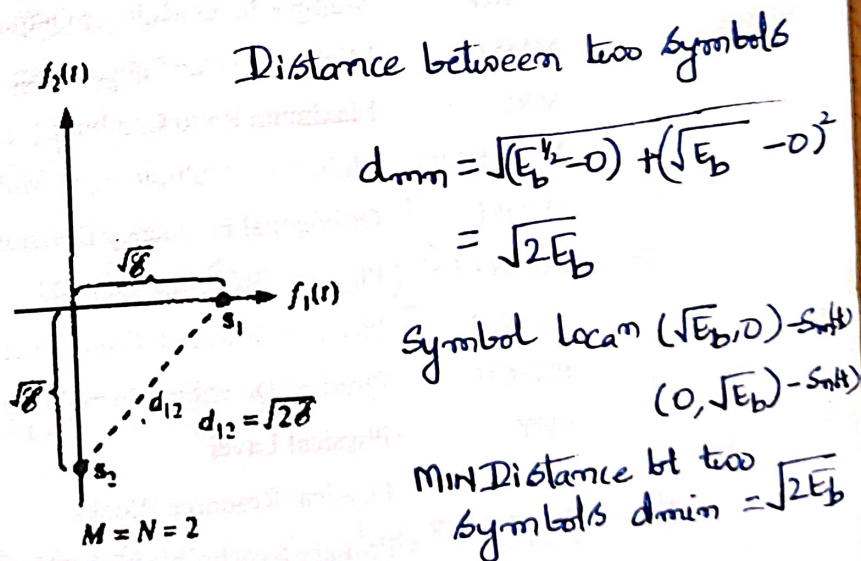
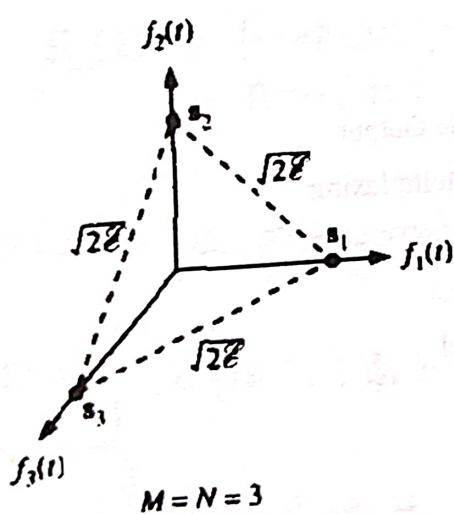
$$\langle s_m(t), s_n(t) \rangle = \begin{cases} E & \text{when } m=n \\ 0 & \text{when } m \neq n. \end{cases} \quad 1 \leq m, n \leq M.$$

A set of M orthogonal waveforms $s_m(t)$ has Equal energy and differ in frequency.

Signal Representation

$$s_m(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t + 2\pi m \Delta f t)$$

$$\begin{aligned} \text{Energy } s_m(t) &= \int_0^T \frac{2E_b}{T_b} \cos^2(2\pi f_c t + 2\pi m \Delta f t) dt \\ &= E_b \end{aligned}$$



Distance between two symbols

$$\begin{aligned} d_{mn} &= \sqrt{(\sqrt{E_b} - 0)^2 + (0 - \sqrt{E_b})^2} \\ &= \sqrt{2E_b} \end{aligned}$$

Symbol locam $(\sqrt{E_b}, 0) - s_m(t)$
 $(0, \sqrt{E_b}) - s_n(t)$

MIN Distance bt two symbols $d_{min} = \sqrt{2E_b}$

$$\text{Basis function } \phi_m(t) = \frac{s_m(t)}{\sqrt{E_b}} = \sqrt{\frac{2}{T_b}} \cos(2\pi f_c t + 2\pi m \Delta f t)$$

General Representation of basis for $\phi_m(t) = \frac{s_m(t)}{\sqrt{E_b}}$

As there are M basis function the Vector Representation is

Vector notation

$$S_1 = (\sqrt{E_b}, 0, \dots, 0)$$

$$S_2 = (0, \sqrt{E_b}, \dots, 0)$$

$\vdots$

$$S_m = (0, 0, \dots, \sqrt{E_b})$$

Symbol Energy $E_s = K E_b$

$$E_b = \frac{E_s}{\log_2 M}$$

$$\text{Also } d_{\min} = \sqrt{2E_s} = \sqrt{2E_b \log_2 M}$$

CONDITION FOR ORTHOGONALITY

$$S_1(t) = A \cos(2\pi f_1 t + 2\pi \eta \Delta t) \text{ rep. it as } A \cos(2\pi f_1 t + \phi_1)$$

$$S_2(t) = A \cos(2\pi f_2 t + 2\pi \eta \Delta t) \\ = A \cos(2\pi f_2 t + \phi_2)$$

Condition for orthogonality is take cross $\int_0^T S_1(t) S_2(t) dt = 0$

$$0 = \int_0^T \cos(2\pi f_1 t + \phi_1) \cdot \cos(2\pi f_2 t + \phi_2) dt$$

$$= \frac{1}{2} \int_0^T \cos(2\pi(f_1 + f_2)t + (\phi_1 + \phi_2)) dt + \cos(2\pi(f_1 - f_2)t + (\phi_1 - \phi_2)) dt$$

$$= \int_0^T \cos 2\pi(f_1 + f_2)t \cdot \cos(\phi_1 + \phi_2) - \sin 2\pi(f_1 + f_2)t \sin(\phi_1 + \phi_2) dt$$

$$+ \int_0^T \cos 2\pi(f_1 - f_2)t \cdot \cos(\phi_1 - \phi_2) - \sin 2\pi(f_1 - f_2)t \sin(\phi_1 - \phi_2) dt$$

Integrate and solve to get

$$0 = \cos(\phi_1 + \phi_2) \frac{\sin \frac{2\pi(f_1 + f_2)T}{2\pi(f_1 + f_2)}}{\frac{2\pi(f_1 + f_2)}{2\pi(f_1 + f_2)}} + \sin(\phi_1 + \phi_2) \left[\frac{\cos \frac{2\pi(f_1 + f_2)T}{2\pi(f_1 + f_2)} - 1}{\frac{2\pi(f_1 + f_2)}{2\pi(f_1 + f_2)}} \right] \\ + \cos(\phi_1 - \phi_2) \frac{\sin \frac{2\pi(f_1 - f_2)T}{2\pi(f_1 - f_2)}}{\frac{2\pi(f_1 - f_2)}{2\pi(f_1 - f_2)}} + \sin(\phi_1 - \phi_2) \left[\frac{\cos \frac{2\pi(f_1 - f_2)T}{2\pi(f_1 - f_2)} - 1}{\frac{2\pi(f_1 - f_2)}{2\pi(f_1 - f_2)}} \right]$$

When $\phi_1 \neq \phi_2$

$$2\pi(f_1 + f_2)T = 2m\pi$$

$$m = 1, 2, \dots$$

$$2\pi(f_1 - f_2)T = 2n\pi$$

$$n = 1, 2, \dots$$

$$f_1 + f_2 = m/T_b$$

$$f_1 - f_2 = n/T_b$$

$$\Delta f = f_1 - f_2 = n/T_b$$

When $\phi_1 = \phi_2 = \phi$

$$0 = \cos 2\phi \sin \frac{2\pi(f_1 + f_2)T}{2\pi(f_1 + f_2)} + \sin 2\phi \left[\cos \frac{2\pi(f_1 + f_2)T}{2\pi(f_1 + f_2)} - 1 \right] \\ + \sin \frac{2\pi(f_1 - f_2)T}{2\pi(f_1 - f_2)}$$

$$\text{then } 2\pi(f_1 + f_2)T_b = 2m\pi$$

$$m = 1, 2, \dots$$

$$2\pi(f_1 - f_2)T_b = n\pi$$

$$n = 1, 2, \dots$$

$$\Delta f = f_1 - f_2 = \frac{n}{2T_b}$$

$$\Delta f_{\min} = 1/2T_b$$

To achieve orthogonality the adjacent frequency slot $\Delta f = 1/2T_b$ represents the min. frequency separation between signals.

FREQUENCY SHIFT KEYING.

$$S_m(t) = \text{Re} \{ A_m g(t) e^{j2\pi f_c t} \} \text{ where } t \in [0, T_b]$$

$$A_m = e^{j2\pi m \Delta f}$$

$$g(t) = \sqrt{\frac{2E_b}{T_b}}$$

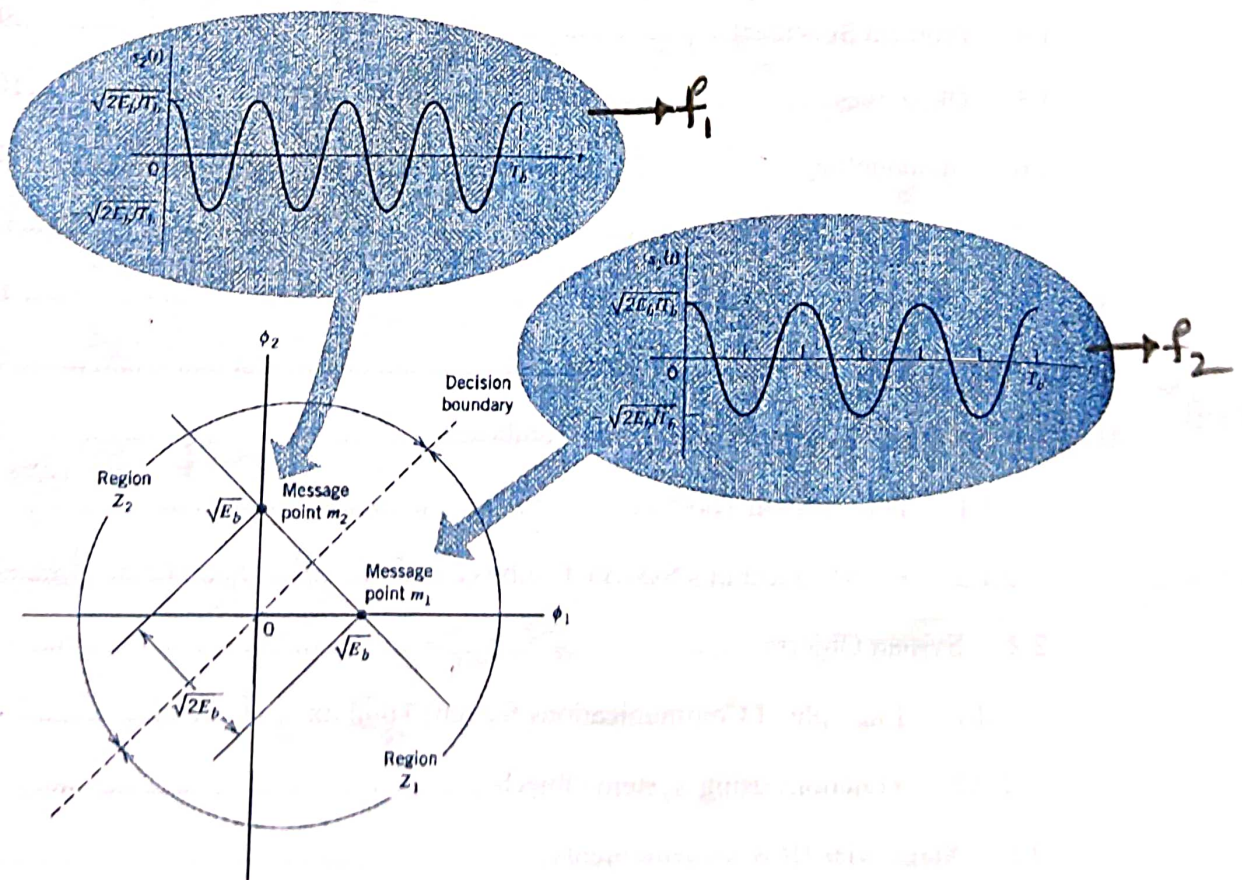
$$S_m(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t + 2\pi m \Delta f t)$$

In Binary Frequency Shift Keying for signal $S_1(t)$ and $S_2(t)$ to be orthogonal $\Delta f = f_1 - f_2 = 1/2T_b$

Then

$$S_1(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t + \frac{\pi h}{T_b} t) = \sqrt{\frac{2E_b}{T_b}} \cos 2\pi f_1 t$$

$$S_2(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t - \frac{\pi h}{T_b} t) = \sqrt{\frac{2E_b}{T_b}} \cos 2\pi f_2 t$$



Energy per bit = $\int_0^{T_b} S_1^2(t) dt$

= $\int_0^{T_b} \frac{2E_b}{T_b} \cos^2 2\pi f_1 t dt$

= E_b

Energy of both signals are same

Basis function $\phi_1(t) = \frac{S_1(t)}{\sqrt{E_b}} = \sqrt{\frac{2}{T_b}} \cos 2\pi f_1 t$

Basis function $\phi_2(t) = \sqrt{\frac{2}{T_b}} \cos 2\pi f_2 t$

Since FSK orthogonal signals all basis functions have only 'i' component. ie cos.

Signal Representation

$S_1(t) = \sqrt{E_b} \phi_1(t)$ 0 to T_b

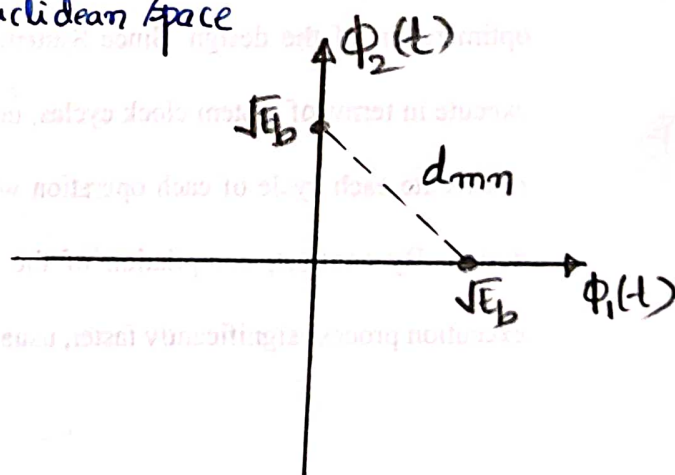
$S_2(t) = \sqrt{E_b} \phi_2(t)$ 0 to T_b

Symbol representation in Euclidean space

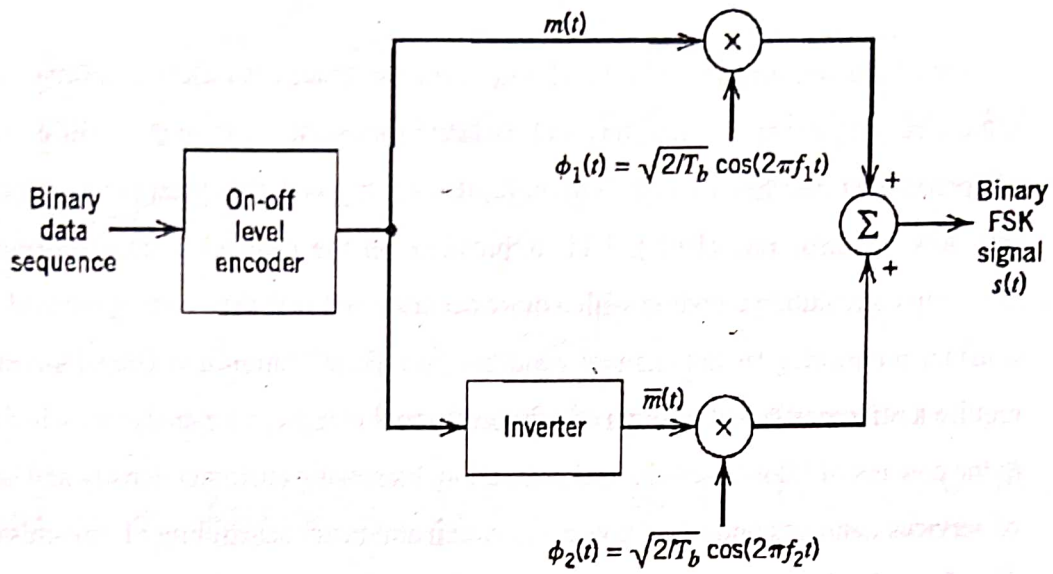
$S_1(t) = (\sqrt{E_b}, 0)$

$S_2(t) = (0, \sqrt{E_b})$

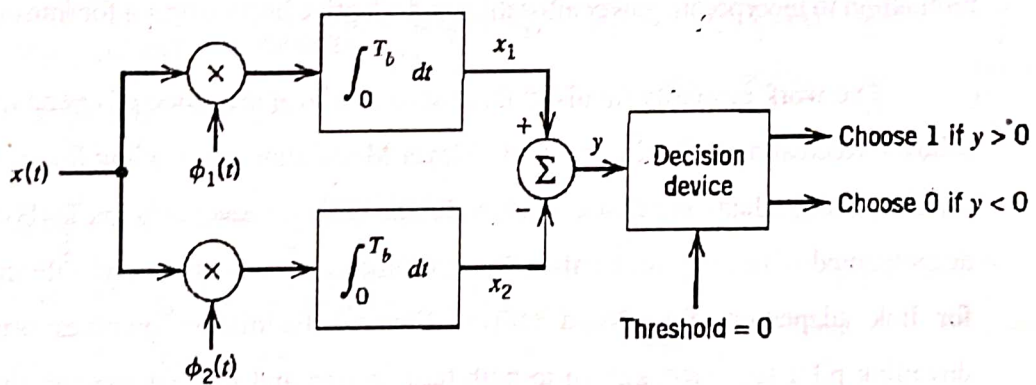
$d_{min} = \sqrt{2E_b}$



Symbol Energy $E_s = K E_b$
 $= \log_2 E_b$
 $E_s = E_b$



(a) FSK MODULATION



FSK DEMODULATION

MINIMUM SHIFT KEYING

- Use of efficient spectral modulation techniques
- Attributes of MSK are constant envelope, spectral efficiency, error rate performance of binary PSK and self synchronizing capability

In bandlimited channels spectrally efficient modulation techniques is used to save bandwidth

- Objective of spectrally efficient modulation is to maximize bandwidth efficiency. defined as -
ratio of data rate to channel BW (bits/s/Hz)
- (ii) These mod schemes has to achieve BW efficiency at prescribed av. bit error rate with min signal power

The signal ~~mod~~ of MSK is defined as

$$s(t) = \sqrt{\frac{2E_b}{T_b}} \cos \theta(t) \cos 2\pi f_c t - \sqrt{\frac{2E_b}{T_b}} \sin \theta(t) \sin 2\pi f_c t$$

Further : 1. It has constant envelope

~~s(t)~~ 2. There is phase continuity in RF carrier at bit transition

3. The signal is an FSK signal with signalling freq

$$f_+ = f_c + \frac{1}{4T} ; f_- = f_c - \frac{1}{4T} \therefore \Delta f = 1/2T$$

4. This is min freq spacing which allows the 2 FSK signals to be coherently orthogonal.

4. The excess phase of MSK signal to the carrier phase

$$\theta(t) = b_k(t) \frac{\pi t}{2T} = \pm \frac{\pi t}{2T}$$

which increases or decreases linearly during each bit period T

A bit b_k of +1 corresponds to an increase of carrier phase by 90°
-1 ————— || decrease by 90°

The continuous phase frequency shift keying signal is expressed as

$$S(t) = \sqrt{\frac{2E_b}{T_b}} \cos \left[2\pi f_c t + \theta(t) \right] \quad \text{--- (1)}$$

here $\theta(t) = \theta(0) \pm \frac{\pi h}{T_b} t$

Putting $\theta(t) = \theta(0) + \frac{\pi h}{T_b} t$ in (1)

$$S(t) = \sqrt{\frac{2E_b}{T_b}} \cos \left[2\pi f_c t + \theta(0) + \frac{\pi h}{T_b} t \right]$$

$$= \sqrt{\frac{2E_b}{T_b}} \cos \left[2\pi \left(f_c + \frac{h}{2T_b} \right) t + \theta(0) \right] \quad 2(a)$$

Similarly $\theta(t) = \theta(0) - \frac{\pi h}{T_b} t$

$$S(t) = \sqrt{\frac{2E_b}{T_b}} \cos \left[2\pi \left(f_c - \frac{h}{2T_b} \right) t + \theta(0) \right]$$

If comparing with CPFSK, then

$$f_2 = f_c - \frac{h}{2T_b} \quad ; \quad f_1 = f_c + \frac{h}{2T_b}$$

Then $f_c = \frac{f_1 + f_2}{2}$ and $h = T_b(f_1 - f_2)$

h is deviation ratio

For $h = \frac{1}{2}$ The frequency deviation is half the bit rate
ie min freq. spacing that allows FSK signals representing
1 and 0 to be coherently orthogonal

Thus CPFSK with deviation ratio of $\frac{1}{2}$ is
MINIMUM SHIFT KEYING

MSK as OQPSK

$$S(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t + \theta(t))$$

$$= \sqrt{\frac{2E_b}{T_b}} \cos 2\pi f_c t \cos \theta(t) - \sqrt{\frac{2E_b}{T_b}} \sin 2\pi f_c t \sin \theta(t)$$

In phase component

$$S_I(t) = \sqrt{\frac{2E_b}{T_b}} \cos \theta(t) \cos 2\pi f_c t$$

$$= \sqrt{\frac{2E_b}{T_b}} \cos 2\pi f_c t \cos(\theta(0) \pm \frac{\pi}{2T_b} t)$$

$$= \sqrt{\frac{2E_b}{T_b}} \cos \theta(0) \cos \frac{\pi}{2T_b} t \cos 2\pi f_c t$$

$$\begin{matrix} + & \rightarrow & 0 \\ - & \rightarrow & \pi \end{matrix}$$

$$-T_b \leq t \leq T_b$$

Energy of $S_I(t) = \sqrt{E_b}$ for baseband

$$\therefore \phi(t) = \frac{\sqrt{\frac{2E_b}{T_b}} \cos \frac{\pi(t)}{2T_b} \cos 2\pi f_c t}{\sqrt{E_b}} = \sqrt{\frac{2}{T_b}} \cos \frac{\pi t}{2T_b} \cos 2\pi f_c t$$

Quadrature component

$$S_Q(t) = \sqrt{\frac{2E_b}{T_b}} \sin \theta(0) \sin \frac{\pi t}{2T_b} \sin 2\pi f_c t$$

$$\text{Ily } \phi_2(t) = \sqrt{\frac{2}{T_b}} \sin \frac{\pi t}{2T_b} \sin 2\pi f_c t$$

Representation in terms of basis fn

$$S_I(t) = \sqrt{E_b} \cos \theta(0) \phi_1(t) \quad \theta \text{ to } -T_b \text{ to } T_b$$

$$S_Q(t) = \sqrt{E_b} \sin \theta(0) \phi_2(t) \quad 0 \text{ to } 2T_b$$

To determine the phase change

$$S(t) = \sqrt{\frac{2E_b}{T_b}} \cos\left(2\pi f_c t \pm \frac{\pi h t}{T_b} + \theta(0)\right)$$

Represent ± 1 as a_n

$$\frac{t}{T_b} \text{ as } \left(\frac{t - nT_b}{T_b}\right)$$

then

$$S(t) = \sqrt{\frac{2E_b}{T_b}} \cos\left(2\pi f_c t + a_n \pi h \frac{(t - nT_b)}{T_b} + \theta(0)\right]$$

For pulse durⁿ. denote $\frac{(t - nT_b)}{T_b}$ as $g(t - nT_b)$

$$\begin{aligned} \therefore S(t) &= \sqrt{\frac{2E_b}{T_b}} \cos\left(2\pi f_c t + 2\pi h a_n g(t - nT_b) + \theta(0)\right] \\ &= \sqrt{\frac{2E_b}{T_b}} \cos\left(2\pi f_c t + \theta(0) + \phi(t, I)\right) \end{aligned}$$

where

$$\begin{aligned} \phi(t, I) &= 2\pi h a_n g(t - nT_b) \text{ for large no. of symbols} \\ &= 2\pi h \sum_{-\infty}^{\infty} a_n \int_{-\infty}^t g(\tau - nT_b) d\tau \\ &= 2\pi h \sum a_n q(nT) \end{aligned}$$

here

$$q(nT) = \int_{-\infty}^t g(\tau - nT_b) d\tau$$

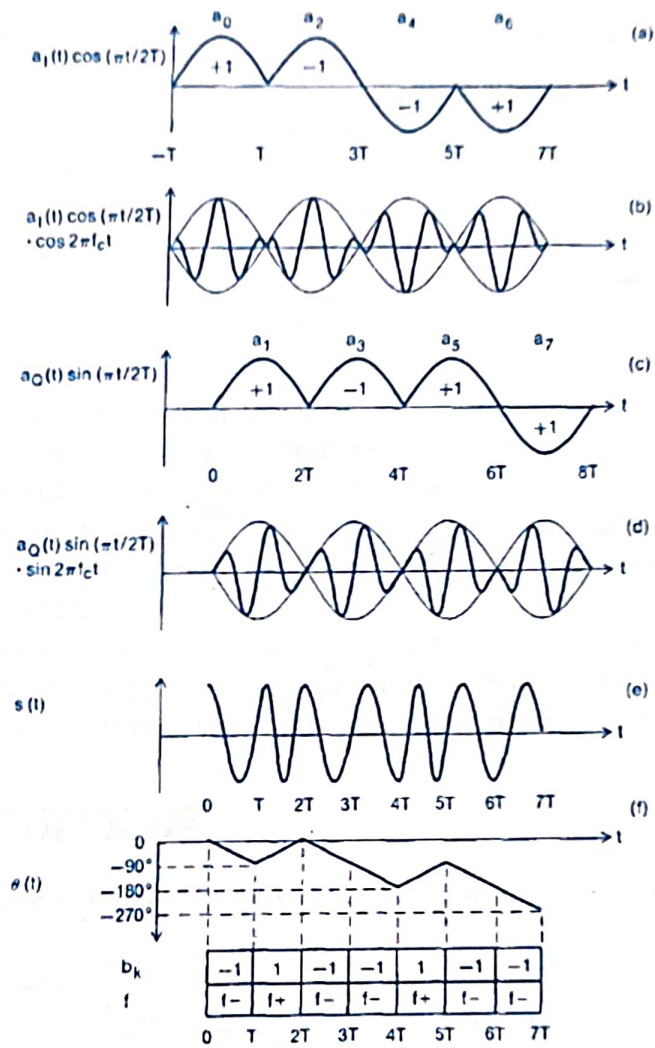


Fig. 5. MSK waveforms.

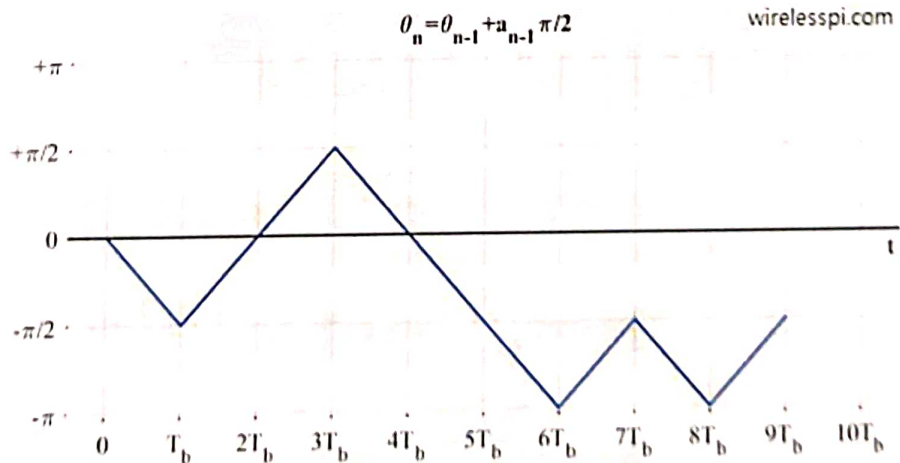
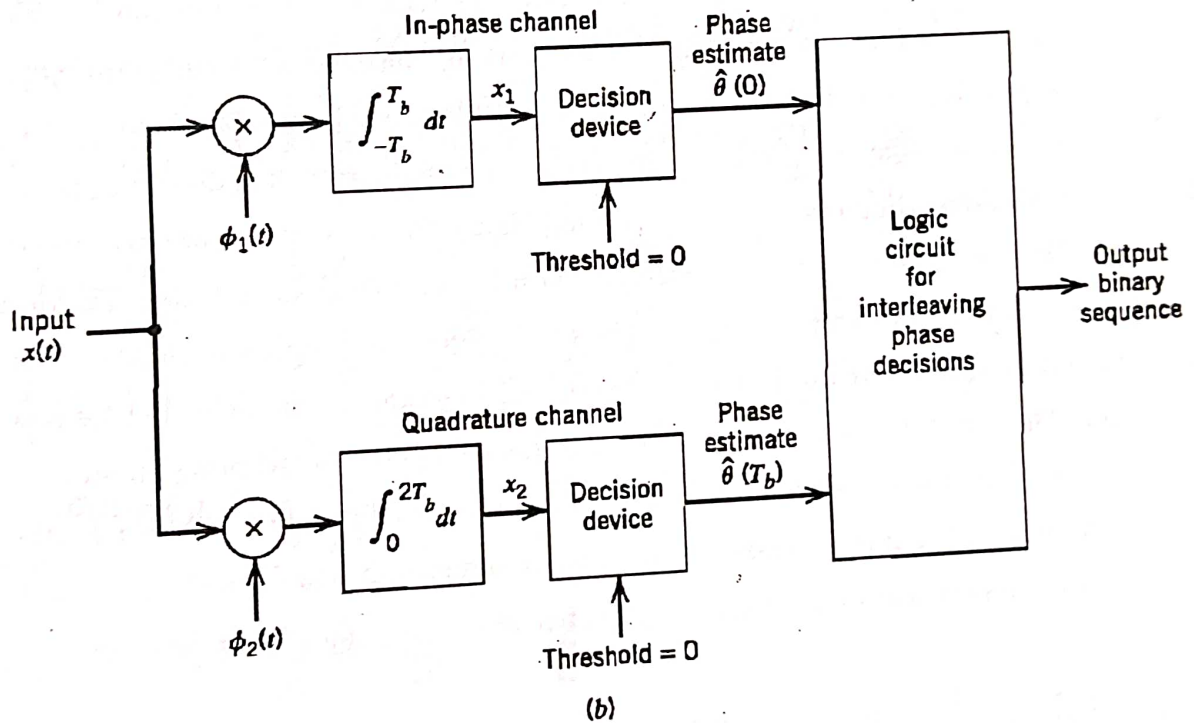
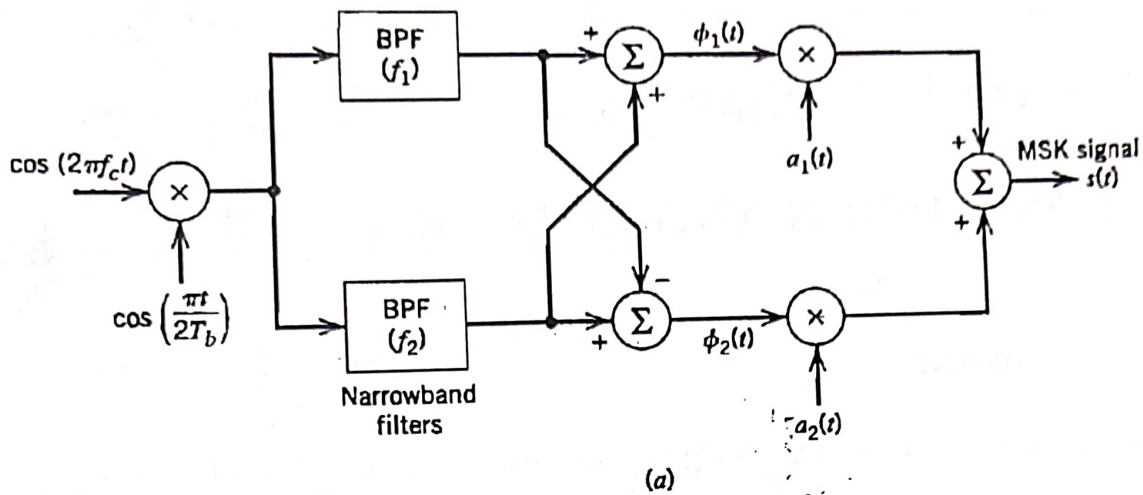


Figure 5: θ_n evolving with time



MSK

PSD of MSK: Baseband

$$S_x(f) = \frac{32 E_b}{\pi^2} \left[\frac{\cos(2\pi T_b f)}{16 T_b^2 f^2 - 1} \right]^2$$

PROBLEMS.

1. Sketch the phase trellis diagram for MSK signal.

Given $g(t) = \frac{1}{4} \frac{1}{2T}$; $h = \frac{1}{2}$ $0 \leq t \leq T$. Find the phases

Soln. $\phi(t, 1) = 2\pi h \sum a_m q(mT)$

$$q(m) = \int_0^T \frac{1}{2T} dt = \frac{1}{2}$$

$$\phi(t, 1) = 2\pi h a_m \frac{1}{2} = 2\pi \left(\frac{1}{2}\right) \left(\frac{1}{2}\right) a_m = \frac{\pi}{2} a_m$$

$a_m \in [1, -1]$: Start from '0', send out all 1's to extract the period

$$0, 0 + \frac{\pi}{2}(1) = \frac{\pi}{2}; \frac{\pi}{2} + \frac{\pi}{2}(1) = \pi; \pi + \frac{\pi}{2}(1) = \frac{3\pi}{2}; \frac{3\pi}{2} + \frac{\pi}{2}(1) = 2\pi \approx 0$$

All phases $(0, \pi/2, \pi, 3\pi/2)$

2. If $h = 2/3$ $g(t) = \frac{1}{2T}$

$$\phi(t, 1) = 2\pi h \sum a_m q(mT)$$

$$q(mT) = \int_0^T \frac{1}{2T} dt = \frac{1}{2}$$

$$\phi(t, 1) = 2\pi \frac{2}{3} a_m \frac{1}{2} = \frac{2\pi}{3} a_m$$

Phases $0, 0 + \frac{2\pi}{3} = \frac{2\pi}{3}, \frac{2\pi}{3} + \frac{2\pi}{3} = \frac{4\pi}{3}, \frac{4\pi}{3} + \frac{2\pi}{3} = 2\pi \approx \text{repeat}$

Phases $(0, \frac{2\pi}{3}, \frac{4\pi}{3})$

3. If $h = 3/4$ $g(t) = \frac{1}{2T}$

$$N.K.T \quad q(mT) = \int_0^T \frac{1}{2T} dt = \frac{1}{2}$$

$$\phi(t, 1) = 2\pi \frac{3}{4} \frac{1}{2} a_m = \frac{3\pi}{4} a_m$$

Phases $0, 0 + \frac{3\pi}{4} = \frac{3\pi}{4}, \frac{3\pi}{4} + \frac{3\pi}{4} = \frac{3\pi}{2}, \frac{3\pi}{2} + \frac{3\pi}{4} = \frac{9\pi}{4} - \frac{2\pi}{4} = \frac{7\pi}{4}, \frac{7\pi}{4} + \frac{3\pi}{4} = 3\pi$

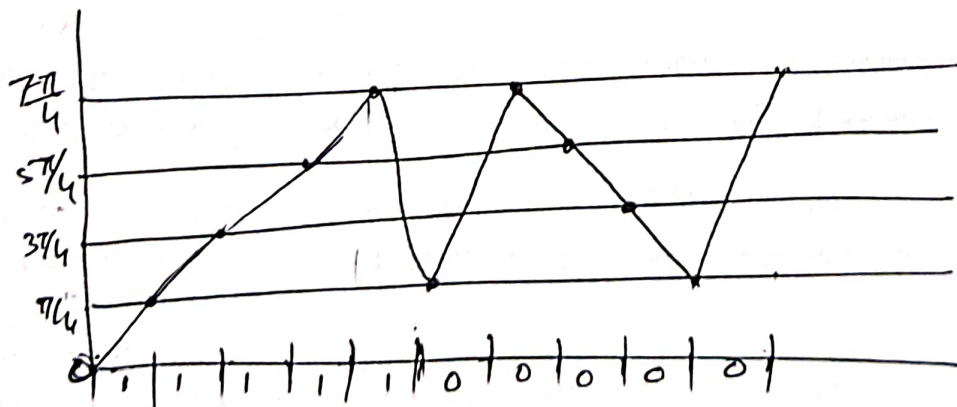
$\{0, \frac{3\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}, \pi, \frac{5\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}\}$ $\pi + \frac{3\pi}{4} = 2\pi$ repeat

Sketch the trellis diagram for the sequence {1111100000} employing partial-response MSK with $h = \frac{1}{2}$ and low pass shaping signal $g(t) = 1/(4T)$ over $[0, 2T]$.

Q1 $h = \frac{1}{2}$ $q(T) = \int_0^T \frac{1}{4T} dt = \frac{1}{4}$, $q(2T) = \int_0^{2T} \frac{1}{4T} dt = \frac{1}{2}$

phase = $2\pi h \sum_{n=0}^L I_n q(nT) = 2\pi \frac{1}{2} (a_{n-1} \frac{1}{4} + a_n \frac{1}{2})$
 $= \pi/4 a_{n-1} + \pi/2 a_n$

symbol	Past ϕ	Added ϕ	Total ϕ	Total ϕ Mod 2π
1	0	$\pi/4$	$\pi/4$	$\pi/4$
1	$\pi/4$	$\pi/2$	$3\pi/4$	$3\pi/4$
1	$3\pi/4$	$\pi/2$	$5\pi/4$	$5\pi/4$
1	$5\pi/4$	$\pi/2$	$7\pi/4$	$7\pi/4$
1	$7\pi/4$	$\pi/2$	$9\pi/4$	$\pi/4$
0	$\pi/4$		$-\pi/4$	$7\pi/4$
0	$7\pi/4$	$-\pi/2$	$5\pi/4$	$5\pi/4$
0	$5\pi/4$	$-\pi/2$	$3\pi/4$	$3\pi/4$
0	$3\pi/4$	$-\pi/2$	$\pi/4$	$\pi/4$
0	$\pi/4$	$-\pi/2$	$-\pi/4$	$7\pi/4$



Bandwidth in M-PSK.

$$B = \frac{2}{T}$$

$$\text{bit rate } R_b = 1/T_b$$

$$m = 2^k$$

$$B = \frac{2R_b}{\log_2 m}$$

$$T = T_b \log_2 m$$

$$\rho \text{ bandwidth efficiency} = \frac{R_b}{B} = \frac{\log_2 m}{2}$$

$$\text{If } m=2 \quad \rho = 0.5$$

$$m=4 \quad \rho = 1$$

$$m=8 \quad \rho = 1.5 \text{ bps/Hz}$$

$$\text{Symbol duration } T_s = T_b \log_2 m \quad 2^k = m$$

Bandwidth of M-FSK

$$B = \frac{m}{2T}$$

$$T = T_b \log_2 m = \frac{\log_2 m}{R_b}$$

$$= \frac{m R_b}{2 \log_2 m}$$

$$\rho = \frac{R_b}{B} = \frac{2 \log_2 m}{\log_2 m}$$

$$\text{If } m=2 \quad \rho = 1$$

$$4 \quad \rho = 1$$

$$8 \quad \rho = 0.75$$

$$16 \quad \rho = 0.5$$

1. Determine tx bandwidth reduction and average signal of 256 QAM compared to 64 QAM.

Problem 6.15

The transmission bandwidth of 256-QAM signal is

$$B = \frac{2R_b}{\log_2 M}$$

where R_b is the bit rate given by $1/T_b$ and $M = 256$. Thus

$$B_{256} = \frac{2(1/T_b)}{\log_2 256} = \frac{2}{16T_b} = \frac{1}{8T_b}$$

The transmission bandwidth of 64-QAM is

$$B_{64} = \frac{2(1/T_b)}{\log_2 64} = \frac{2}{8T_b} = \frac{1}{4T_b}$$

Hence, the bandwidth advantage of 256-QAM over 64-QAM is

$$\frac{1}{4T_b} - \frac{1}{8T_b} = \frac{1}{8T_b}$$

The average energy of 256-QAM signal is

$$\begin{aligned} E_{256} &= \frac{2(M-1)E_0}{3} = \frac{2(256-1)E_0}{3} \\ &= 170E_0 \end{aligned}$$

where E_0 is the energy of the signal with the lowest amplitude. For the 64-QAM signal, we have

$$E_{64} = \frac{2(63)}{3}E_0 = 42E_0$$

Therefore, the increase in average signal energy resulting from the use of 256-QAM over 64-QAM, expressed in dBs, is

$$\begin{aligned} 10 \log_{10} \left(\frac{170E_0}{42E_0} \right) &\approx 10 \log_{10}(4) \\ &= 6 \text{ dB} \end{aligned}$$

Problem-1

A PAM partial-response signal (PRS) is generated as shown in Figure P3.14 by exciting an ideal lowpass filter of bandwidth W by the sequence

$$B_n = I_n + I_{n-1}$$

at a rate $1/T = 2W$ symbols/s. The sequence $\{I_n\}$ consists of binary digits selected independently from the alphabet $\{1, -1\}$ with equal probability. Hence, the filtered signal has the form

$$v(t) = \sum_{n=-\infty}^{\infty} B_n g(t - nT), \quad T = \frac{1}{2W}$$

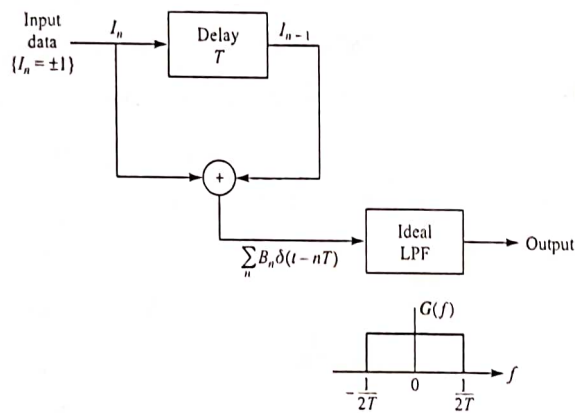


FIGURE P3.14

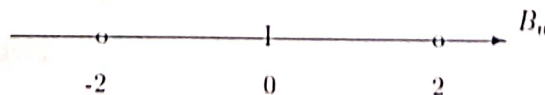
1. Sketch the signal space diagram for $v(t)$, and determine the probability of occurrence of each symbol.
2. Determine the autocorrelation and power density spectrum of the three-level sequence $\{B_n\}$.

Solution:

1. $B_n = I_n + I_{n-1}$. Hence :

I_n	I_{n-1}	B_n
1	1	2
1	-1	0
-1	1	0
-1	-1	-2

The signal space representation is given in the following figure, with $P(B_n = 2) = P(B_n = -2) = 1/4$, $P(B_n = 0) = 1/2$.



2.

$$\begin{aligned} R_{BB}(m) &= E[B_{n+m}B_n] = E[(I_{n+m} + I_{n+m-1})(I_n + I_{n-1})] \\ &= 2R_{ii}(m) + R_{ii}(m-1) + R_{ii}(m+1) \end{aligned}$$

Since the sequence $\{I_n\}$ consists of independent symbols :

$$R_{ii}(m) = \begin{cases} E[I_{n+m}]E[I_n] = 0 \cdot 0 = 0, & m \neq 0 \\ E[I_n^2] = 1, & m = 0 \end{cases}$$

Hence :

$$R_{BB}(m) = \begin{cases} 2, & m = 0 \\ 1, & m = \pm 1 \\ 0, & \text{o.w} \end{cases}$$

and

$$\begin{aligned} S_{BB}(f) &= \sum_{m=-\infty}^{\infty} R_{BB}(m) \exp(-j2\pi f m T) = 2 + \exp(j2\pi f T) + \exp(-j2\pi f T) \\ &= 2[1 + \cos 2\pi f T] = 4 \cos^2 \pi f T \end{aligned}$$

$$E[|G(f)|^2] = (AT)^2 \frac{\sin^2(\pi f T)}{(\pi f T)^2}$$

$$S_{xx}(f) = 4 \frac{(AT)^2}{T} \left(\frac{\sin \pi f T}{\pi f T} \right)^2 \cos^2 \pi f T$$

Sketch the phase trellis diagram for MSK for following
 4. $h = \frac{3}{4}$ ii Seq [1 1 1 0 0] Given $g(t) = \frac{1}{2T}$ for $0 \leq t \leq T$

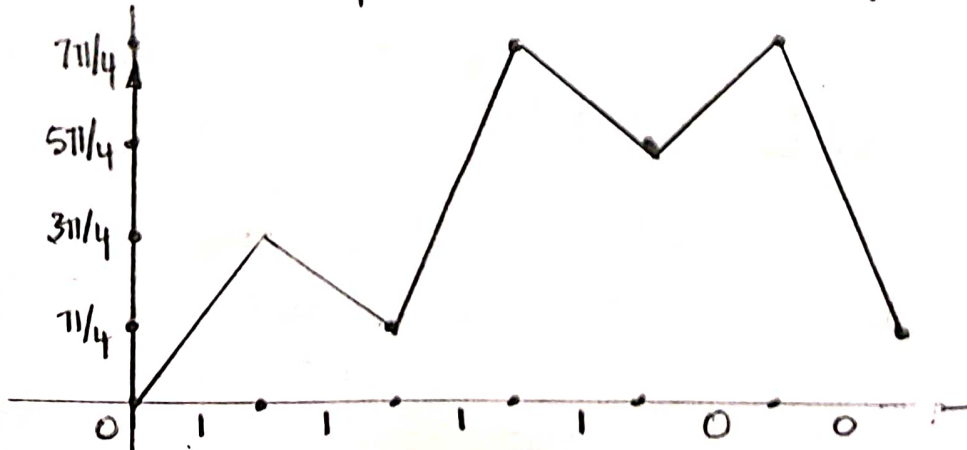
$$g(t) = \frac{1}{2T} \quad g(\tau) = \int_0^T \frac{1}{2T} dt = \frac{1}{2}$$

$$g(2\tau) = \int_0^{2T} \frac{1}{2T} dt = 1$$

$$\text{Phase} = 2\pi \frac{3}{4} \sum_{n=1}^N \left(\frac{a_{n-1}}{2} + a_n \right) = \frac{3\pi}{2} (a_{n-1} + a_n)$$

$$= \frac{3\pi}{4} a_{n-1} + \frac{3\pi}{2} a_n$$

Symbol	Past ϕ	Added ϕ	Total	Mod 2π
1	0	$3\pi/4$	$3\pi/4$	$3\pi/4$
1	$3\pi/4$	$3\pi/2$	$9\pi/4$	$\pi/4$
1	$\pi/4$	$3\pi/2$	$7\pi/4$	$7\pi/4$
1	$7\pi/4$	$3\pi/2$	$13\pi/4$	$5\pi/4$
0	$5\pi/4$	$-3\pi/2$	$-\pi/4$	$7\pi/4$
0	$7\pi/4$	$-3\pi/2$	$\pi/4$	$\pi/4$



2b.

$$I_n = a_n - a_{n-1}$$

$$PSD = \frac{|G(f)|^2}{T} \sum_{m=-\infty}^{\infty} R_{XX} e^{-j2\pi f m T}$$

$$u(t) = \sum_n I_n g(t - nT)$$

$$R_{XX} = E[I_{n+m} I_n]$$

$$= E[(a_{n+m} - a_{n+m-1})(a_n - a_{n-1})]$$

$$R_{XX}(m) = E[a_{n+m}a_n - a_{n+m-1}a_n - a_{n+m}a_{n-1} + a_{n+m-1}a_{n-1}]$$

$$R_{XX}(0) = E[a_n a_n - a_{n-1} a_n - a_n a_{n-1} + a_{n-1} a_{n-1}] = 2$$

$$R_{XX}(1) = E[a_{n+1}a_n - a_n a_n - a_{n+1}a_{n-1} + a_n a_{n-1}] = -1$$

$$R_{XX}(-1) = E[a_{n-1}a_n - a_{n-2}a_n - a_{n-1}a_{n-1} + a_{n-2}a_{n-1}] = -1$$

$$R_{XX}(2) = E[a_{n+2}a_n - a_{n-1}a_n - a_{n+2}a_{n-1} + a_{n+1}a_{n-1}] = 0$$

$$PSD = \frac{|G(f)|^2}{T} [2 - e^{-j2\pi f T} - e^{j2\pi f T}]$$

$$= \frac{|G(f)|^2}{T} [2 - 2\cos 2\pi f T]$$

$$PSD = \frac{4}{T} |G(f)|^2 \sin^2 \pi f T$$

Question: A lowpass equivalent representation of a PAM signal is given by

$$u(t) = \sum I_n g(t - nT) \quad I_n = a_n - a_{n-1}$$

$g(t)$ is rectangular pulse, a_n 's are uncorrelated binary (1, -1) random variables with equal probability

Find R_{XX} , PSD.